Justine Dupal

Human Factors + UX Researcher
Design Strategist + Service Designer

After attaining both a B.S. (Cornell) and M.S. (Harvard) in Human Factors and Ergonomics, I have spent 10+ years using mixed methods research to uncover user needs and translate them into design solutions, spanning industries, products, and services. The versatility of human factors is why I love it so much!

My design thinking process is a mixture of:

- Applied Human Factors
- Design Research
- Systems Thinking
- Quantitative + Qualitative methods

Through my projects, I consider the:

- Employee, customer, and patient experiences
- Modes of service touchpoints
- Platforms, technology, and databases needed
- Capabilities of LLM's and AI to enhance research methods, end-user insights, and potential solutions

The common theme is my **empathy** for the user and being an **advocate for the users' needs.**

	Surveys	Interviews	Focus groups	Ethnographic Research	Workshops	Behavioral Mapping	Journey Mapping	Data Analysis	Process Mapping	Service Blueprints	Wireframes + Prototypes	Assessments + Testing	Other tools
Hospital + Healthcare	✓	✓	✓	✓		✓	✓	✓	✓		Checklists	Human Factors Analysis	Ergonomic Tools
Product Design + UX	✓	✓		✓			✓				Wireframes + Prototypes	Usability Testing	
Architecture + Interiors	✓	✓	✓	✓	✓	✓		✓		Space programming	Computer Simulations	POE Evaluation	Change Management
Pharmaceutical Services		✓	✓		✓	✓	✓	✓	✓	Service Blueprints	Wireframes + Prototypes		
Government Consulting	✓	✓	✓	✓	✓		✓	✓	✓	Service Blueprints	Org Restructure	Usability Scoring	Change Management

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J&J: Designing Hope: A Nurse Navigator Program For Treatment-Resistant Depression

Design Problem:

A novel antidepressant for treatment-resistant depression (TRD) and major depressive disorder (MDD) with suicidal ideation had shown promising results in clinical trials. However, unlike typical antidepressants that are taken orally at home, this medication was different.

J&J had created a ketamine nasal spray, a black box drug requiring administration While traditional anti-depressants are taken orally in a daily dose, this new treatment is administered nasally, under **direct supervision of a nurse, 1-2 times week**. Patient must be **monitored for side-effects for 2 hours** after dosing.

Patient Challenges:

This care model creates significant barriers for patients who need the treatment the most:

- **Care Coordination:** Patient must leave home, requiring childcare, pet care, transportation, time off work, etc.
- **Post-Dose Experience:** 2-hour monitoring means side effect + anxiety management
- **Financial Complexity:** Insurance reimbursement process unclear+ difficult
- **Access Limitations:** Only REMS-certified facilities can offer treatment
    - Limited availability, especially for rural patients

Early adoption revealed a crisis: **patient drop off's despite drug's efficacy**

Program Challenges:

- High Drop-off Rate
- Insurance Uncertainty
- Barriers to Access
- Operational Burdens
- Low Referral

Further Considerations:

- **Business Challenge: Scalability:** # patients 16K → 200K first year, reach ~360K by 2025
    - Without addressing operational barriers + patient support gaps, growth impossible
- **Operational expansion: Translatability:** in-office to home health option next year
- **Patient Experience: Technological Integration:** adaptable, seamless shift to app

My Role:
Senior Service Design Strategist on a 4-person design team (Lead Designer, Copywriter, Junior Designer) in partnership with J&J's clinical and commercial teams. Over 10 months, as our team designed and launched the nurse navigator support program, I was responsible for patient journey + process mapping, all call guides, digital messaging, journal revisions, review committee submissions, and app strategy. I led meetings with other contracting agencies on website design and served as an overall patient experience advocate.

Goal:
Create a scalable patient support ecosystem that reduces dropout, addresses operational burdens, increases referrals, and maintains compliance with FDA regulations for a vulnerable mental health population, while simultaneously preparing for future home health delivery and future mobile application.

Solution:

A nurse support program was designed to help patients onboard the treatment.

- **Supportive calls, texts, and therapeutic resources will be delivered during crucial parts of patient's journey**
- Nurse navigator assists patients with:
    - Setting expectations
    - Providing educational and medical info
    - Sharing resources on drug and its administration
- Home health option + phone / tablet application

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Anti-depressant Nurse Support Program

Approach: HCD Under Regulatory Constraints

Added complication: In order to design our comprehensive nurse navigator program, we had to ensure complete compliance with FDA / legal / medical review teams at every step of the process, as any word said to the patient had to be approved. We had to build in time for these review processes and justify our design decisions to the regulation committees with extensive annotations and evidence.

PHASE 1:

Population Data

Patient Interviews

Examine existing data
Create personas and map patient journey; identify pain points requiring most support and intervention

Best Practice Research

Previous Pilot Data

Apply other learnings
Utilize insights from similar programs and pilot study on mental health nurse navigators

PHASE 2:

Client Workshops

Stakeholder Interviews

Evaluate progress
Meet with stakeholders, vendors, legal / FDA teams, and other contract teams often to mold program and remain aligned

Patient Workshop

Focus Groups

User test program before launch
Sample patient population to assess, challenge and validate program design

Process: HCD process (7 months)

Human-centered design calls for an examination of users and their journeys, while reiterating and improving the design as pain points are revealed. Limited resources and time to produce deliverables means continuous conversations with stakeholders on the short-term and long-term strategy, what to expect now vs. later, and when each rollout is realistic.

Who are the users?

Patients, Caregivers, HCP / Office Staff, Program Staff

What is the patient's journey like?

Journey Mapping

Process Mapping

How can our service help pain points?

Resource Allocation

Maximizing Impact

What needs to be addressed now vs. later?

Prioritizing Deliverables

How do we measure our success?

KPI & Growth Metrics

Data Collection

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Anti-depressant Nurse Support Program

Who are the users? | What is the patient's journey like? | How can our service help pain points? | What needs to be addressed now vs. later? | How do we measure our success?

Patients

Demographics: men & women 18+ with TRD, MDD, or MDSI

- Ethnicity, language, & finances vary
- 18-29 yrs old and women more commonly diagnosed with MDD
- ~80% commercially-insured

Location: within United States

Cultural Considerations: Racial and cultural differences affect if patient seeks care + care relevance / success

Special Considerations: patients only prescribed if treatment-resistant to ≥2 meds or have suicide ideation

Attributes:
Cynical - been through many different treatments, so wary of a new one
Want honesty – no sugar-coating
Modern - may prefer texting over calls

Caregivers

Relation: can be a legal guardian, authorized romantic partner, family member, or friend

Responsibilities may include:

- Patient transportation
- Pet and/or childcare during apt
- Assist patient in finding financial options for treatment
- May act as patient liaison

Special Considerations: some cultures or relationships not as likely to emotionally support MDD males

Attributes:
Concerned – want to know what to expect and how they can support
Accommodating - many rearrange their lives for patient's treatment

HCP / Office Staff

Location: facilities will need to be REMS trained & certified, limiting availability of eligible offices

Special Considerations:

- Private or group treatment areas?
- Language matches patient?
- Culturally diverse to ensure patients feel represented?
- Home Health: safety of navigator in patient's home

Attributes:
Busy - may not have a lot of room for the office visits this requires, may not be super responsive
Crowded - some have created group treatment sessions to accommodate patient demand

Navigators + Case Managers

Demographic: background in healthcare, especially mental health

Location: Remote

Special Considerations:

- Legibility and useability of materials and dashboard
- Program simple enough for multiple patients per navigator
- Training – content and length?

Attributes:
Empathetic – come from backgrounds of patient support and want to do their best to help
Limited – handling many patients at once through complicated processes

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Anti-depressant Nurse Support Program

Who are the users? | **What is the patient's journey like?** | How can our service help pain points? | What needs to be addressed now vs. later? | How do we measure our success?

➢ What are the biggest issues for patients?

Assessing existing research: Population data, research, and past interviews revealed important insights into the patient experience

Patient Feedback: Background data illuminated patients' general feedback with the current treatment process

Patient Likes:

+ Rapid onset of action & durable response
+ Perceived safety of on-site monitoring
+ Sense of community
+ Ability to co-schedule counseling

Patient Dislikes:

- Loss of time
- Lack of privacy / comfort at centers
- Unreliable providers
- Side effects
- Need for transportation / childcare
- Reimbursement hurdles

Patient Insight Themes: I synthesized research into 6 core patient needs that would guide all design decisions:

1. **Anxiety Reduction:** "Make this as simple, authentic and empathetic as possible"
2. **Trust:** "I need to believe this will work and you're being honest with me"
3. **Preparation:** "Tell me what to expect. Don't sugarcoat, but don't scare me"
4. **Control:** "I feel powerless; give me agency where you can"
5. **Acknowledgement:** "See me as a person, not just a patient"
6. **Respectful Candor:** "Be authentic, not clinical"

These became our design principles for every patient touchpoint.

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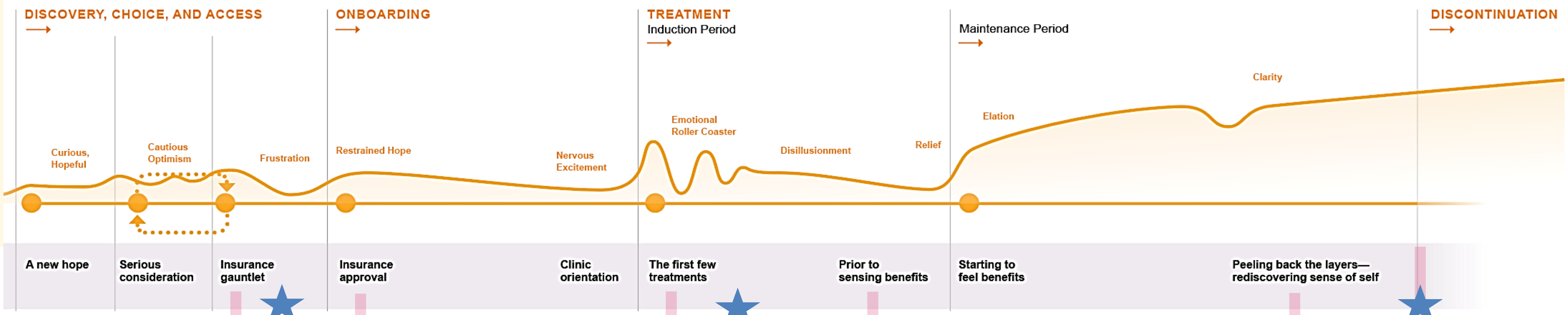
Anti-depressant Nurse Support Program

Who are the users? | **What is the patient's journey like?** | How can our service help pain points? | What needs to be addressed now vs. later? | How do we measure our success?

- **Key Strategic Decision:** Journey Map of complete patient experience exposed **pain points where support is needed** the most, especially during the **early months**
- **Dropout** clustered around specific moments as patients endure a **bumpy ride** before noticing relief from symptoms

Journey Map: Major events and emotions were mapped out to discern high's and low's of patient's treatment journey:



Inconsistent onboarding across clinics and treatment centers

Confusing language used to describe medical terms and treatment processes

Emotional roller coaster during first few treatments, as side effects and experiences vary by patient

Frustrating induction process from cycles of altered states of mind, exhaustion, and life interruption

Adding other therapy, like music, journaling, meditation, coaching, etc., are sampled

Unclear expectations as prescriber identifies long-term dosage and treatment cadence and patient wonders about long-term reliance on drug

Critical Dropout Moments:

1. **Pre-treatment** (after prescription, before first dose): Insurance uncertainty, logistical overwhelm
2. **Weeks 1-4** (induction): Side effects, emotional rollercoaster, unclear expectations
3. **Weeks 8-12** (maintenance transition): Confusion about long-term treatment plan

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Anti-depressant Nurse Support Program

Who are the users? | **What is the patient's journey like?** | How can our service help pain points? | What needs to be addressed now vs. later? | How do we measure our success?

➢ **Patient Barriers discovered**

BARRIER 1: The Logistical Gauntlet

Treatment required patients to:
- Find a REMS-certified clinic (limited availability)
- Arrange transportation twice weekly
- Secure childcare / pet care for 3+ hours per appointment
- Take time off work during business hours
- Sit in shared treatment rooms with strangers during vulnerable moments

For a patient already struggling with depression and low motivation, each logistical hurdle compounded the obstacles.

BARRIER 2: Insurance Uncertainty

The drug was expensive (~$800-$1,000 per dose). Reimbursement processes were complex. Patients faced:
- Prior authorization delays
- Unexpected denials
- Out-of-pocket costs they couldn't afford
- Confusion about coverage

Many patients stopped treatment not because it didn't work, but because they couldn't navigate the insurance maze.

BARRIER 3: Emotional Rollercoaster

During first 4-8 weeks (induction phase), patients experienced:
- Varying side effects, like dissociation, nausea, dizziness
- Exhaustion from frequent appointments
- Frustration with interrupted daily life
- Doubt about whether it was working
- Anxiety over how long they will have to continue this process for treatment to stay effective

Logistics have a mental and emotional toll on patient, so how can our service make it easier?

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Anti-depressant Nurse Support Program

Who are the users? — What is the patient's journey like? — **How can our service help pain point?** — What needs to be addressed now vs. later? — How do we measure our success?

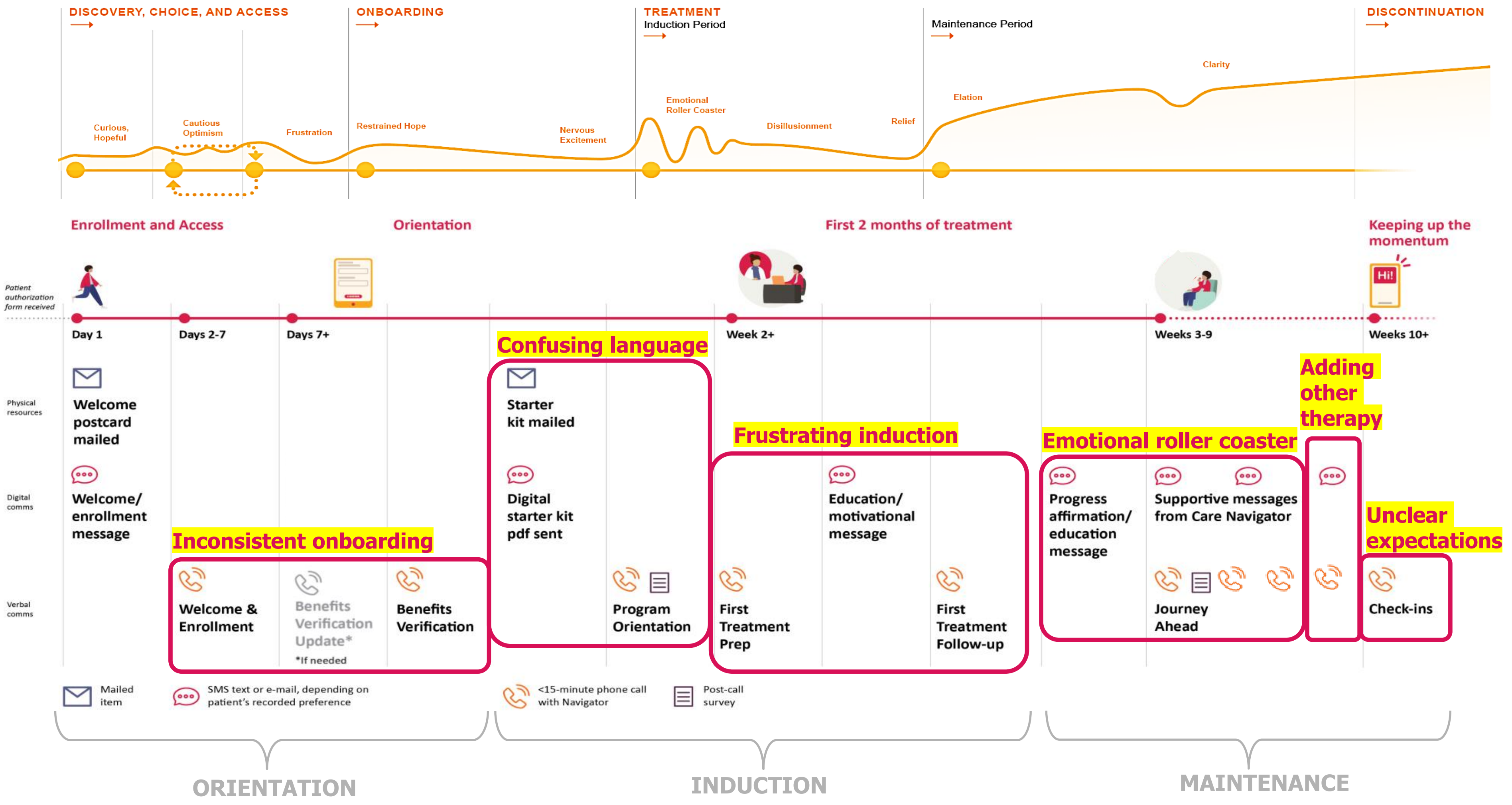
PHASE 2: Service Design & Touchpoint Strategy

- Pain points led to solution mapping, informing where to place touchpoints in our supportive ecosystem
- **Each set of calls / messages / resources** was designed to **address a main pain point**
  - **When** patient hears from navigator
  - **What** navigator should discuss
  - **Medium** of message

ORIENTATION PERIOD (Pre-dose to Week 2):
Touchpoint: Orientation calls
Goal: Set expectations, build trust, answer questions
Content: Treatment process, side effects, insurance navigation, scheduling

INDUCTION PHASE (Weeks 2-9):
Touchpoint: Weekly check-in calls + text messages
Goal: Normalize experiences, provide encouragement, troubleshoot issues
Content: "What you're feeling is normal," emotional support, appointment reminders

MAINTENANCE TRANSITION (Weeks 10+):
Touchpoint: Bi-monthly calls + on-demand resources
Goal: Help patient understand long-term treatment plan
Content: Dosage adjustments, sustainability, integrating other therapies



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Anti-depressant Nurse Support Program

Who are the users?

What is the patient's journey like?

How can our service help pain point?

What needs to be addressed now vs. later?

How do we measure our success?

➢ The Deliverables: A Comprehensive Support Ecosystem - Passive and interactive resources prioritized according to budget, time, and necessity

Immediate Impact:

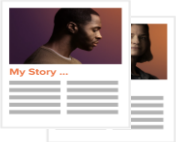
- **Remote Services + Mailings –** keep patients informed to decrease drop-off rate, operational burden, and insurance uncertainty
- **Call + Reference Guides** – prepared content to guide navigators in patient interactions
- **Website + Online Resources** - provide tools, assistance, and navigation to patients and caregivers throughout treatment
- **Texts/E-mails** – deliver program information to patients and caregivers
- **Journal + Reflective Activities –** aid patients incrementally and holistically throughout treatment journey

PHASE I

Trail Guide [1]

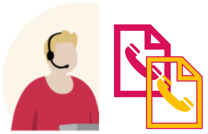
Starter Kit [2]

Resources + Financial Options [3]



Peer Stories [4]

Nurse Navigator Training Materials [5]



Nurse Navigator Call Guides [6]

Text + E-mail Messages [7]

Website

Digital Starter Kit [8]

Reflection Journal + Goal Trackers [9]

1. ***Trail Guide:*** *Treatment handbook of what patient can expect over the next 2 years*
2. ***Starter Kit:*** *Mailed package containing welcome materials, printed resources, website information, stress ball, and other marketing materials*
3. ***Patient Resources****: Drug information, treatment overview. Financial assistance, Website with resource library*
4. ***Peer Story Library:*** *Video testimonials from patients at different treatment stages, Written stories highlighting diverse experiences*
5. ***Training Materials:*** *Navigator training deck (clinical protocols, communication best practices), Cultural competency guidance , Crisis escalation protocols*
6. ***Nurse Navigator Call Guides:*** *Orientation call, Benefits Calls, weekly Check-ins, additional support calls , Crisis support protocol , Maintenance transition guide*
7. ***Digital Communications:*** *50+ text/email templates mapped to journey moments - Appointment reminders, educational content, encouragement messages*
8. ***Digital Starter Kit:*** *Online version accessible via website for immediate access*
9. ***Redesigned Reflection Journal:*** *Updated mood and sleep trackers, Goal-setting prompts, Therapy integration activities*

Future Impact:

In-person services (home health) + virtual resources (app) creation begun but put on hold to complete in later phases

PHASE II

Home Health Delivery Services

(strategy developed, implementation pending)

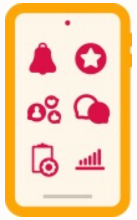
- Visiting nurse administration at patient's home
- Safety protocols for navigators
- Expanded access for rural/mobility-limited patients

PHASE III

Phone App + Orientation

Features prioritized based on patient input:

- Treatment schedule, reminders, alerts
- Secure messaging with navigator
- PHQ-9 mood surveys and progress reports
- Peer stories and community connection
- Educational resources



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Anti-depressant Nurse Support Program

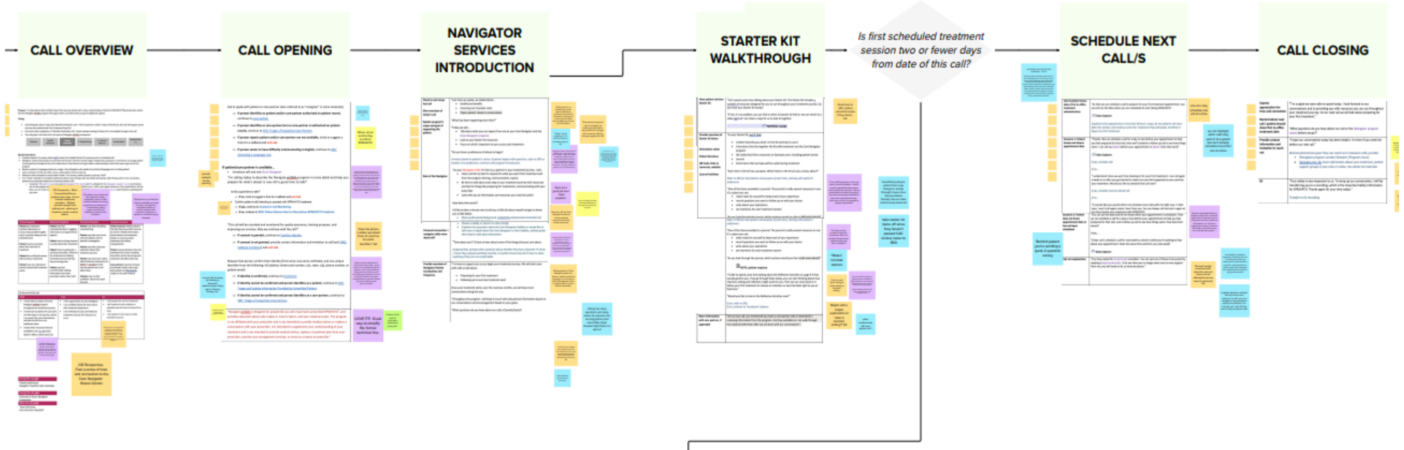
Who are the users? | What is the patient's journey like? | **How can our service help pain point?** | What needs to be addressed now vs. later? | How do we measure our success?

Call Guides

- **Crafted for each key part of journey**
- **Reviewed in weekly team meetings** to assess direction, cadence, and content
- **Addressed:**

Insurance Access & Enrollment | Treatment Side Effects | Supportive resources | Appointment Logistics | Language & Cultural Barriers

ORIENTATION CALL GUIDE



1ST TREATMENT PREPARATION CALL GUIDE

Call Guide Review Process:

Because of population vulnerability, drug class, and legal policies, every word had to be approved in patient-facing material

1. **Internal team review**
2. **Partner + companion agency reviews**
3. **Advisory committee review**
4. **Internal team edits**
5. **Partner edits**
6. **Advisory committee re-review**
7. **Submit to vendor for training decks**

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Anti-depressant Nurse Support Program

Who are the users?

What is the patient's journey like?

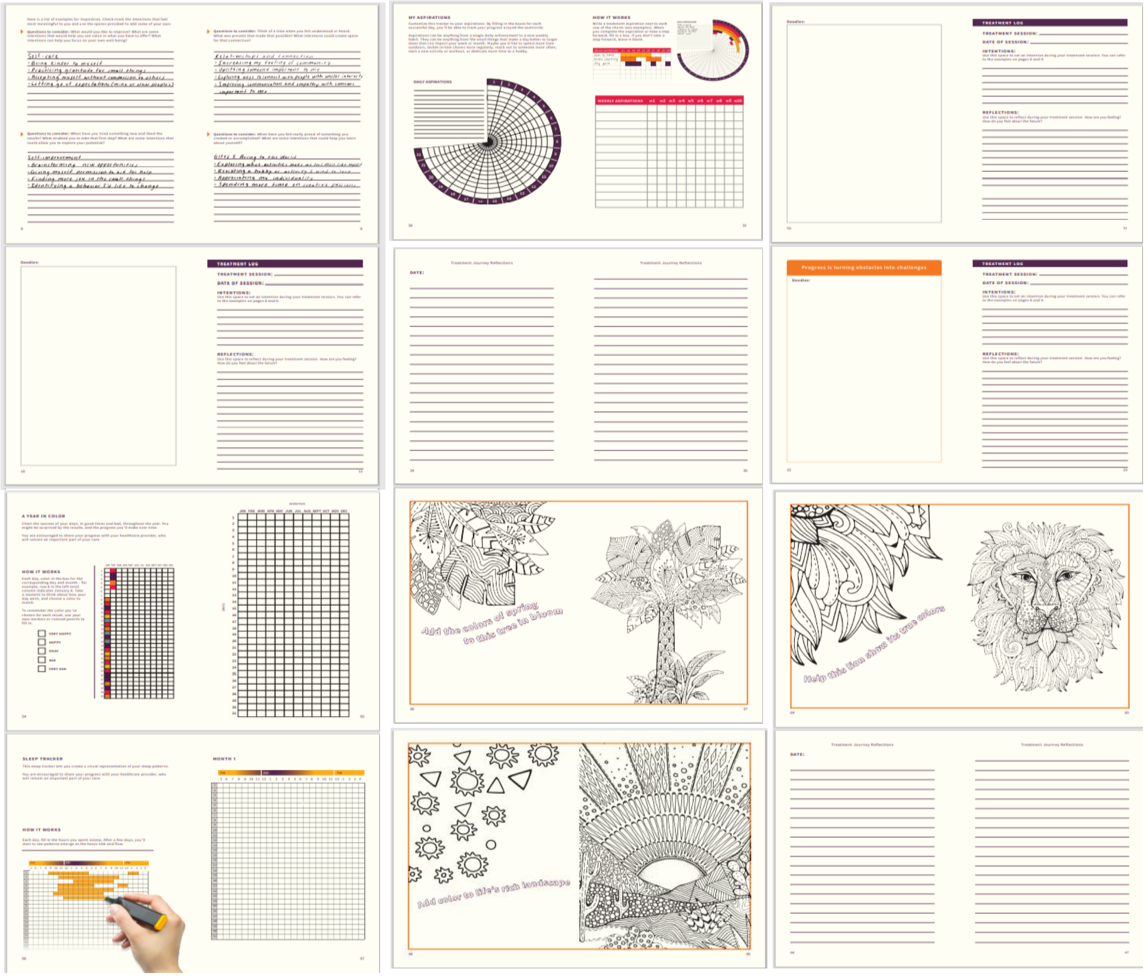
How can our service help pain point?

What needs to be addressed now vs. later?

How do we measure our success?

Starter Kit Updates

- For other support, deliverables needed to be updated, including Starter Kit and Journal pages
- **Working with a partner agency, we re-designed elements of the Journal**
  - Introduced a new activity to assist during drug treatment
  - Updated sleep and mood trackers to help the patient self-monitor their progress during treatment journey
  - Updated branding and coloring designs
- **Upgraded other kit elements,** including "premium" product samples, and added additional brochure of patient services
- **Website updated** (for projected launch in between phase I and phase II) and **virtual Starter Kit** link added to site



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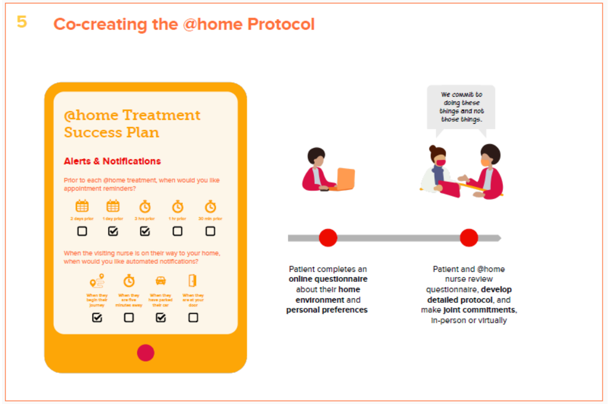
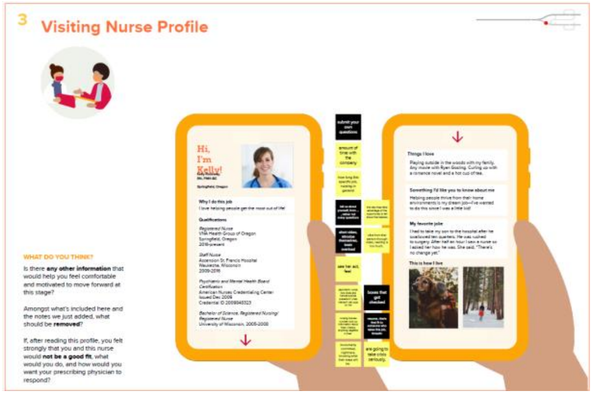
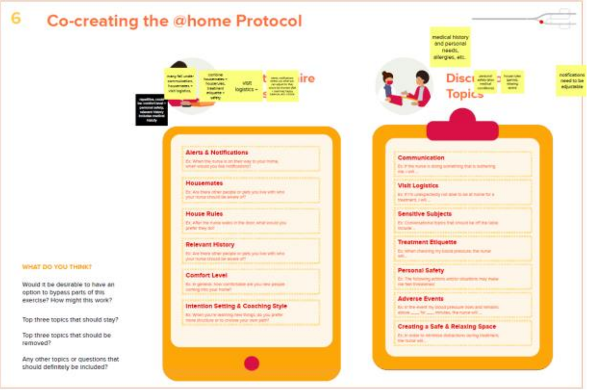
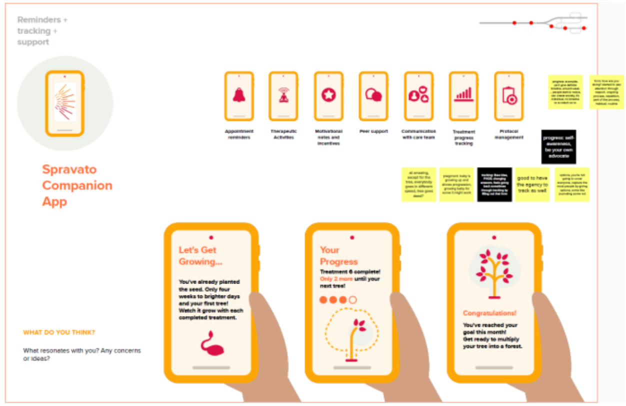
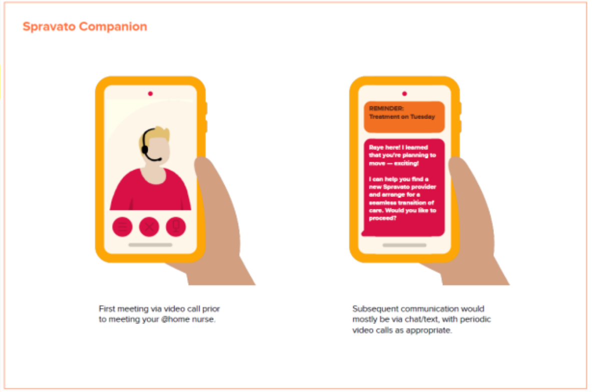
App design and research (put on hold for Phase III):

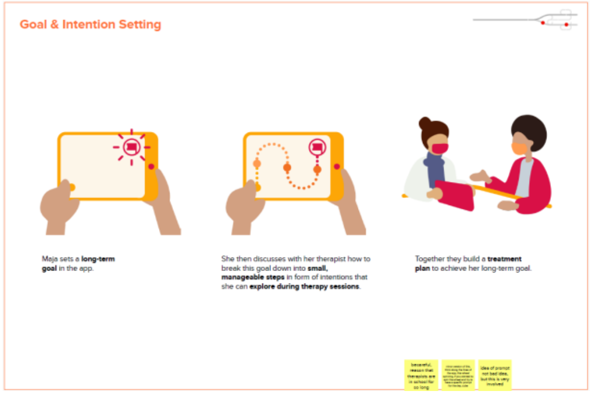












Features requested by patients:

- Treatment schedule, reminders, and alerts
- Communication amongst team
- Peer stories and peer mentors
- Journey Navigation messaging
- PHQ-9 surveys and reports

Other potential desirable features:

- Follow-up surveys and complaint forms
- Prompts for intention setting
- Shipment tracking
- Visiting Nurse proximity alerts
- White noise / relaxing music / guided meditation

Anti-depressant Nurse Support Program

Impact on Patient Experience:

PROJECTED SHORT-TERM IMPACT (6 months post-launch):

- **Dropout reduction:** Streamlined scheduling, educational resources, and care coordination support reduce logistical burden to improve treatment adherence
- **Navigator satisfaction:** Centralized tools, clear protocols, and experience with empathetic support enable navigators to work more efficiently and provide high-value interactions
- **Insurance navigation:** Educational materials and insurance phone call should reduce patient's time spent navigating reimbursement and decrease coverage confusion
- **Referral increase:** Improved patient experience and reduced logistical barriers increase provider confidence in referring

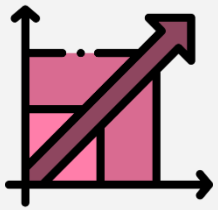
PROJECTED LONG-TERM IMPACT:

- **Support infrastructure** for scaling from 16,000 to 200,000 patients in Year 1
- **Foundation for home health expansion**: enabling access for patients who could not reach clinics before
- **Reduction in operational burden on clinics:** navigators handle logistics so providers can focus on clinical care
- **Improved patient outcomes** through sustained treatment adherence

"My navigator checked in every week during the hardest part. She asked about my kids, my insurance, whether I needed help rescheduling. She made me feel like someone was on my team."

- *Program participant, 8 weeks into treatment*

Business Impact:



Scalability: Program designed to support 10x patient growth with proportional navigator staffing

Regulatory compliance: 100% of materials approved through FDA / legal / medical review process

Cross-functional alignment: Weekly stakeholder reviews ensured clinical, commercial, and legal teams stayed aligned

Vendor readiness: Call guides and training materials enabled smooth handoff to implementation vendor

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Anti-depressant Nurse Support Program

What I Learned:

Designing for Vulnerability
Patients are in crisis, skeptical, and exhausted. It was imperative we designed for the patient’s emotional state instead of just focusing on the process. Our journey map highlights hope, doubt, fear, and relief instead of steps.

Dealing with Regulations
Legal and medical review processes were labor intensive. Quickly I learned what reviewers look for, effective annotations, start early, and have the "why" behind each design choices ready.

Phased Rollout as Strategic Advantage
Launching with core support (calls, resources, website) let us learn from real patients and iterate. Phase II journal redesign wouldn't have been as good without Phase I patient feedback. Perfect is the enemy of launch.

Journey Mapping for Stakeholder Alignment
The teams approached our deliverables with different priorities (adherence, legal risk, growth). Our journey map was a useful unifying visual to present drop out moments that motivated our design choices. Ex: "Patients drop out at Week 4 because of this exact uncertainty, removing this information increases that risk.“

What I'd Do Differently:

Engage Caregivers More Deeply
We created caregiver personas and included them in resources but didn't interview them separately. Caregivers play a huge role in patient adherence (transportation, emotional support), and their needs may differ from patients'. Dedicated caregiver research could have revealed unique support opportunities.

Build Navigator Feedback Loop from Day One
We trained navigators and handed-off call guides but didn't establish a formal feedback mechanism for them to report what was/wasn't working. A routine navigator roundtable could surface issues faster and enable continuous improvement.

Document Cultural Competency Gaps More Explicitly
While patient feedback highlighted language and cultural barriers, this wasn’t a formal research stream. If we had researched more extensively cultural differences in mental health stigma, treatment preferences, and communication styles, it would have made the program more inclusive.

This project taught me that service design in healthcare may at first glance seem to focus on financials and efficiency, but should ultimately be about giving dignity back to the patient. Each touchpoint with the patient was so critically reviewed because it was an opportunity to show support and that their struggle is valid. The nurse navigator program helps remove barriers, provides support, and makes treatment feel possible to population facing a lot of uncertainty.

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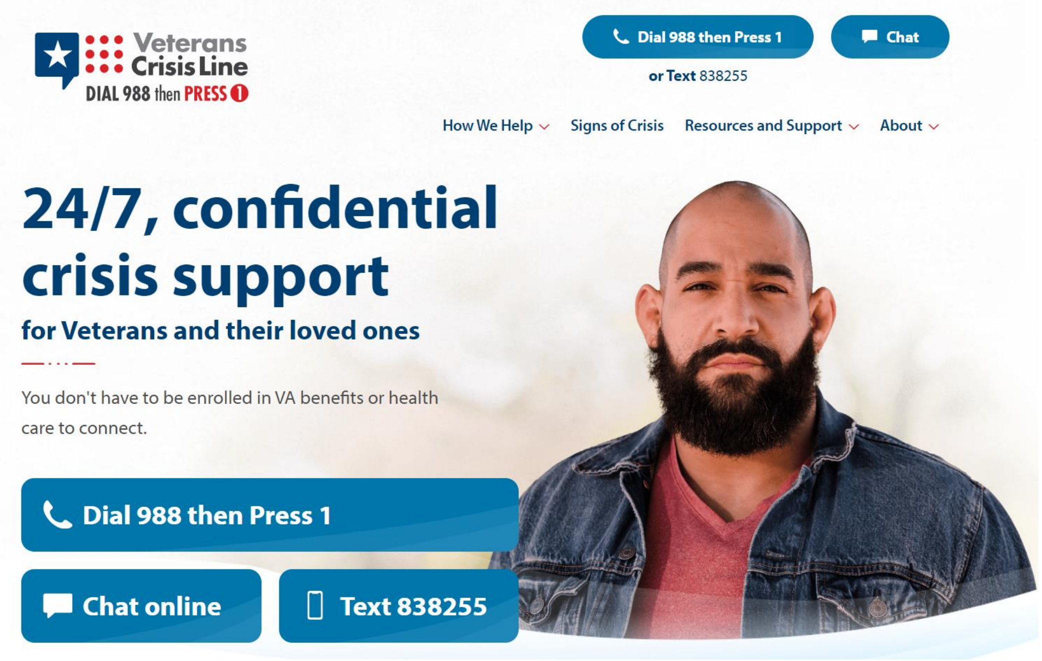
Veterans Crisis Line: Modernizing Mental Health Infrastructure

Design Problem:

Every year, the **Veterans Crisis Line (VCL)** responds to over 900,000 calls, texts, and chats from veterans in mental health crisis, dispatching emergency services 1/25 times. **The White House had recently updated their mandate of "answer every veteran call" with a new goal: answer 90% of VCL calls within 30 seconds.**

However, VCL's legacy Avaya system, non-integrated apps, and Cisco on-prem UCCEX were not future-focused or ready for growth. With increasing call volumes and regulatory requirements, the VA needed to select a new FedRAMP-certified cloud contact center platform (CCaaS) that could scale from 2,000 to 3,500 agents while maintaining the service quality veterans deserve.

My role: Research Lead on a 7-person team tasked with guiding VA's $15M technology decision through the user-centered evaluation of three enterprise vendors (Cisco, Genesys, NICE), as well as the subsequent 5-year implementation and integration strategy of this new software at VCL.



Challenges:

- Technology reliability
- Software and process redundancy
- Dashboard UX
- Routing Inefficiency
- Increase case processing + filing time
- Training remote team

Thie technology procurement + service transformation project affects multiple user groups' needs:

- **Responders** - intuitive dashboards to quickly assess caller risk and connect them to resources
- **Supervisors** - real-time monitoring ("barging") to support responders handling traumatic calls
- **Coordinators** - efficient routing to filter non-crisis calls and reduce responder burnout
- **IT** - FedRAMP compliance, system integration, and infrastructure that could scale 75% in 5 years

VA OIT had strict technical requirements and were funding the project, but the people actually using the system daily (responders et al.) had no seat at the procurement table. **How could we ensure the vendor selection reflected both technical compliance AND frontline needs?**

Data utilized:

- Best Practice Research
- Interviews
- Journey Mapping
- Process Mapping
- Demo / prototype testing

Primary issues identified:

- Support Entry Points
- Dashboards & Reporting
- 'Prank Call' (CWCN) Routing
- Transfers
- Training (Barging)

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Veterans Crisis Line: Modernizing Mental Health Infrastructure

The Approach: Democratizing Technical Procurement

A co-evaluation process would give frontline users equal voice in vendor selection

Phase 1: Understanding the Ecosystem

User Interviews

Field Notes

1. Collect User Data
Canandaigua, NY contact center:

- 31 interviews across 8 user roles (responders, supervisors, coordinators, analysts, managers)
- 34 field observations
- 1 participatory design session

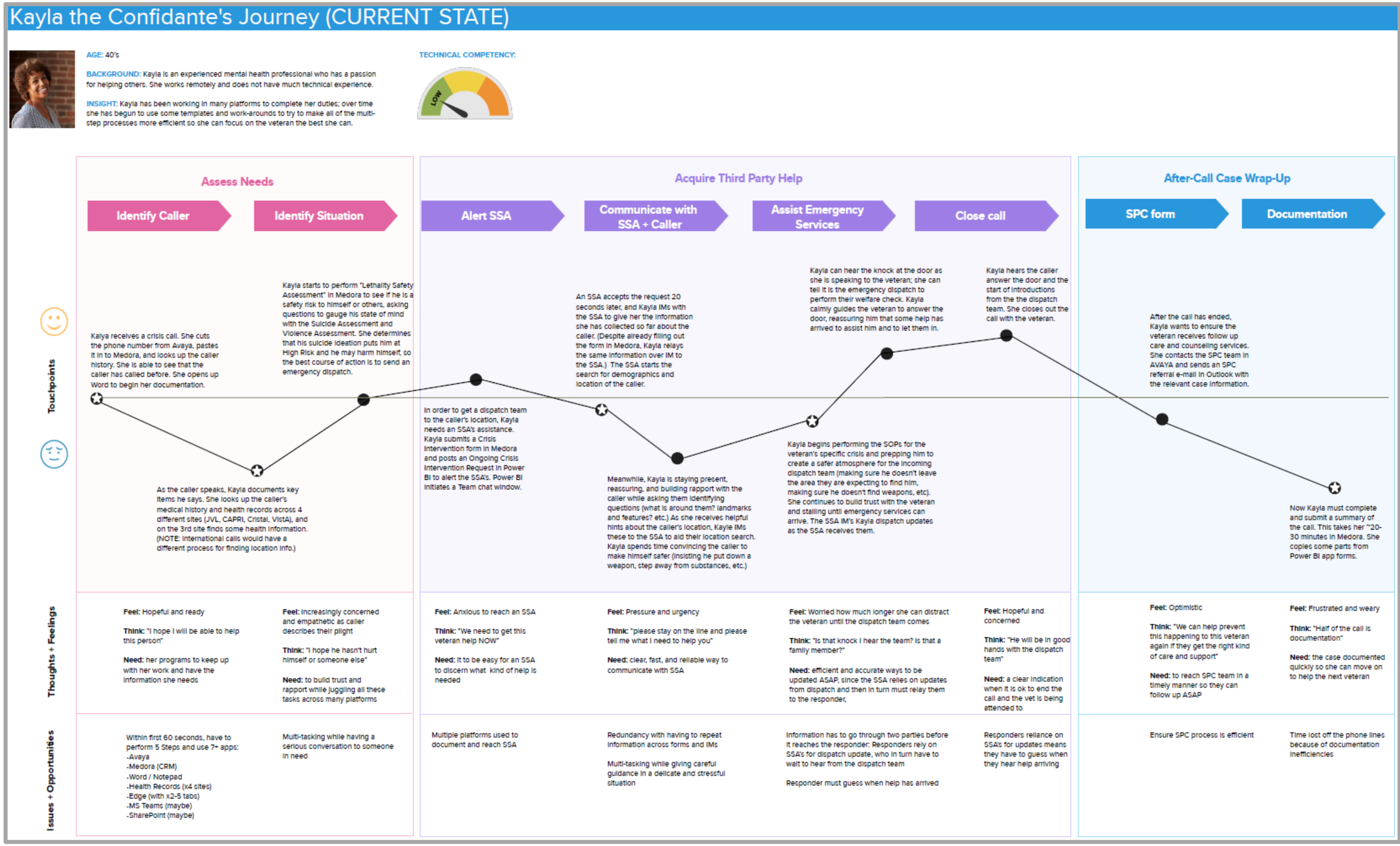
Journey Mapping

Process Mapping

2. Identify Pain Points
Journey mapping and process mapping to identify pain points, communication points, and integration needs

- Personas highlighted the most impacted user groups and their daily experiences at VCL assisting veterans

Key Insights: Users struggled with bad UX and were constantly compensating for **system failures** through **workarounds** that **increased cognitive load** during already distressing interactions. Responders described "**tab chaos**" from toggling between 5-7 disconnected applications mid-call.



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Veterans Crisis Line: Modernizing Mental Health Infrastructure

Phase 2: Translating Research into Testable Criteria

Personas

Use-Cases

Data Analysis and Synthesis, User Personas + Use-Case Creation

I synthesized 100+ hours of research into:
- 12 personas representing primary and secondary users
- 8 "future state" requirements that balanced user needs + technical compliance
- 18 use-case scenarios for 5 call center functions (routing, dashboards, QA, workforce management, agent tools)

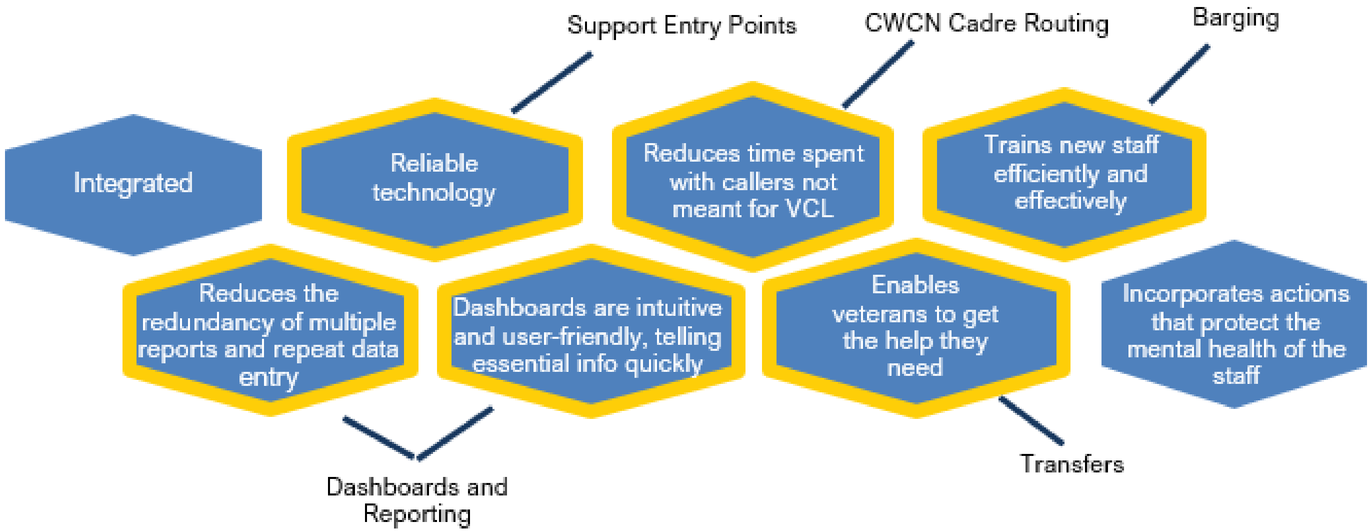
To humanize abstract requirements, we asked vendors to demo realistic scenarios to simulate authentic software interactions and reveal the actual interface VCL staff would use

1. Current State Analysis

VCL Contact Center, a critical call center within VA:
- Aide emergencies
- Connect Vets / loved ones with social services
- Refer Vets to local Suicide Prevention Coordinator (SPC) for counseling and services

Since launch, VCL has supported:
- 7.6 million+ calls
- 360,000+ texts
- 910,000+ chats
- 1.4 million+ referrals to VA SPCs
- 313,000+ emergency service dispatches
- **Overall, connect with >900,000 Veterans a year**

2. Based on data gathered, the ideal future state had 8 features:



3. Key use-cases and subtasks to test main future state priorities, across 5 main categories of call center functions:

Dashboard, Reporting, and Analytics	Call Routing	Agent Management	Quality Assurance	Workforce Optimization and Management
• Dashboard configs	• Transfers • Extensions • Contact presets	• CWCW Cadre Routing • Attribute-based routing / changes • Skill routing / changes	• Barging	
• Set-up process • Real-time data management • Historical data management	• Incoming call notifications • Caller ID		• QA holds • Automated QA	• Shift bids • Capacity / Long-range planning

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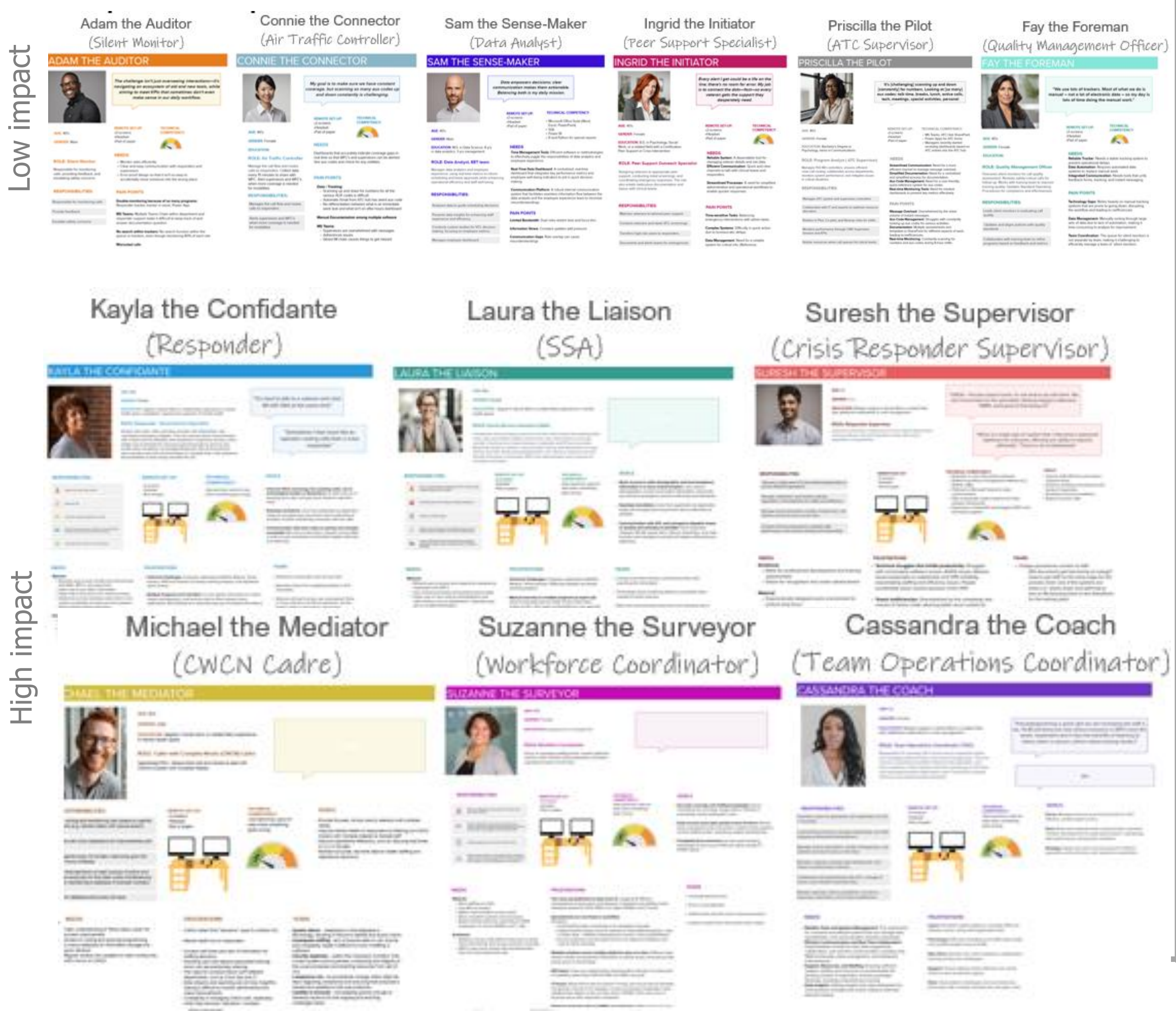
Phase 2: Translating Research into Testable Criteria

4. VCL user groups were defined by job role and then sorted by their likelihood of impact by modernization efforts.



Persona: Responder

KAYLA THE CONFIDANTE

AGE: 40's

GENDER: Female

EDUCATION: degree in Social Work or related field, experience in mental health space, empathetic, experienced supporter of mental health

ROLE: Responder / Social Service Specialist

Answer crisis calls, chats, and texts, provide crisis intervention, and coordinates emergency dispatch. This role requires direct communication with a Social Service Assistant and sometimes a supervisor during a call to initiate and orchestrate the necessary police/emergency services and handle other escalations accordingly. Responder may need to perform warm transfers and cold call third parties to complete their crisis assistance. Documentation is done during and after the call.

"It's hard to talk to a veteran and chat IM with SSA at the same time"

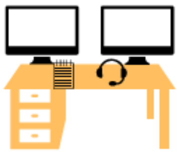
"Sometimes I feel more like an operator routing calls than a crisis responder."

RESPONSIBILITIES:

- Answer crisis calls, chats and texts
- Document calls
- Coordinate emergent dispatches as needed
- Provide crisis interventions to Veterans, Service Members, and their families and submit Suicide Prevention
- Coordinator (SPC) requests to local VA facilities

REMOTE SET-UP:

- 2-screens
- Headset
- Pad of paper



TECHNICAL COMPETENCY:

- Non-technical, asks for help when something goes wrong

GOALS

- **Channel effort and energy into assisting caller, not on technological hurdles or distractions:** be able to focus on assisting each caller and give them whatever help they need
- **Seamless transitions:** move from application to application easily; fill out paperwork around the calls as efficiently as possible; all while maintaining connection with the caller
- **Communication with team mates as quickly and smoothy as possible:** IMs sent to supervisors, requests sent to SSA's, e-mails to local coordinators and facilties happen efficiently and effectively

NEEDS

Material:

- Seamless way to work remote and communicate with SSA's, WFC's, and supervisors
- Faster way to see caller's information
- Faster way to document call / submit summary
- Helpful but concise training to allow them to set up their workstation at home and have answers without always asking supervisors

Emotional:

- Feel like they can trust their equipment and software during these very important calls
- Can communicate with team to debrief after a call as needed; have access to respite to protect their own personal mental health

FRUSTRATIONS

- **Technical Challenges:** Computer, applications (AVAYA, Medora, Teams backup, VPN) and headset not always working properly; chat algorithms seem uneven.
- **Multiple Programs from Call Start:** To even gather information on caller's history, demographics, and services local to them requires many applications. (No dashboard or automatic pop-ups of scraped information.)
- **Inconsistent Communication:** How and when supervisors answer instant messages (IM) is unpredictable; chat responders Teams message to phone responders for outbound SPC calls can be missed among message volume.
- **Call Documentation:** Process (both during and after) is cumbersome, confusing (especially for new hires), and involves manual processes and switching between multiple applications and repositories (Avaya, Avaya CMS, Medora, Cristal (health records), SharePoint, MS Teams, Power BI, etc.) Some consider the apps for documentation not intuitive to use.
- **Volume of Misrouted Calls:** especially on Mondays, responders spend a lot of time routing calls instead of delivering supportive services. Callers trying to contact someone with h a "7" in their extension are often misrouted to the VCL (basic questions / consult appointments, not for VCL)
- **Spam Filters**: Caller ID on many people's devices label the VCL number as "spam" or is one the veterans don't recognize, so they do not pick up.

FEARS

- Hitting the wrong button and can't go back
- Spending critical time navigating desktop to find information
- Veterans will wait to long or go unanswered: Policy is "every veteran's call will be answered": but the longer it takes to document a call, the less responders in the queue available to answer calls, potentially increasing caller wait time
- Not knowing how to access information needed for a Veteran.

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Veterans Crisis Line: Modernizing Mental Health Infrastructure

Phase 3: Co-Creating the Evaluation

User Training

Scorecards

Scorecard Creation + Training

OIT leads decisions in VA culture, so to give end-users a voice within our constraints, we:

- Recruited "Power Users" from Crisis Ops, Quality, Training, and Tech teams
- Trained them on System Usability Scale (SUS) evaluation methodology
- Created parallel scorecards: VCL users scored front-end usability (18 tasks), OIT scored back-end compliance

This gave frontline staff quantifiable data to advocate for their needs in procurement discussions.

Phase 4: Vendor Demonstration & Analysis

Demo / Sandbox testing

Scorecard Results

Three Vendor Demos:

- 3-hour live demos where each vendor performed all 18 tasks
- Real-time scoring by both user and IT groups
- Post-demo workshops to discuss score discrepancies and surface concerns

VCL 'Power Users' assessed front-end usability while IT assessed technical / back-end features

VCL Scorecards

		I'd like using this frequently	System is unnecessarily complex	System is easy to use	I'd need assistance to use this	System functions are well-integrated	Too much inconsistency in system	Most people could learn this system quickly	System is very cumbersome.	I'd feel very confident using this	I'd need to learn a lot of things before using
	DASHBOARD, REPORTING, + ANALYTICS										
SCENARIO	Dashboards for Responders										
	Dashboards for Supervisor										
	Dashboards for Executives										
TASKS	Set-Up Process										
	Real-Time Data										
	Historical Data										
	CALL ROUTING										
SCENARIO	Transfers										
	Extensions										
	Contact Pre-sets / Phonebook										
TASKS	Incoming Call Notifications										
	Caller ID										
	AGENT MANAGEMENT										
SCENARIO	Attribute-Based Routing										
	Skill changes / routing										
	QUALITY ASSURANCE										
S	Barging										
TASKS	QA Holds										
	Automated Holds										
	WORKFORCE OPTIMIZATION + MANAGEMENT										
TASKS	Shift Bids										
	Capacity / Long-range planning										
	Average Score										

Vendor Software Demonstration	Score
Aligns with Business Requirements	
Ease of Use	
Flexibility	
Improved Capabilities	
Overall Impression	
Average Score	

- Both groups assessed the same overall categories
- Both groups used a 5-point scale
- Only VCL users scored subtasks individually
- System Usability Scale was used by 'Power Users' to score each of the 18 tasks during the Demo

	DASHBOARD, REPORTING, + ANALYTICS
SCENARIO	Dashboards for Responders
	Dashboards for Supervisor
	Dashboards for Executives
TASKS	Set-Up Process
	Real-Time Data
	Historical Data
	CALL ROUTING
SCENARIO	Transfers
	Extensions
	Contact Pre-sets / Phonebook
TASKS	Incoming Call Notifications
	Caller ID
	AGENT MANAGEMENT
SCENARIO	Attribute-Based Routing
	Skill changes / routing
	QUALITY ASSURANCE
S	Barging
TASKS	QA Holds
	Automated Holds
	WORKFORCE OPTIMIZATION + MANAGEMENT
TASKS	Shift Bids
	Capacity / Long-range planning

1. I think that I would like to use this system frequently
2. I found the system unnecessarily complex
3. I thought the system was easy to use
4. I think that I would need the support of a technical person to be able to use this system
5. I found the various functions in this system were well integrated
6. I thought there was too much inconsistency in this system
7. I would imagine that most people would learn to use this system very quickly
8. I found the system very cumbersome to use
9. I felt very confident using the system
10. I needed to learn a lot of things before I could get going with this system

OIT Scorecard

5 = Excellent, 4 = Good, 3 = Okay, 2 = Poor, 1 = Worst Imaginable

Aligns with Contact Center Requirements Score:	
Reasoning:	
Ease of Use Score:	
Reasoning:	
Flexibility Score:	
Reasoning:	
Improved Capabilities Score:	
Reasoning:	
Overall Impression Score:	
Reasoning:	

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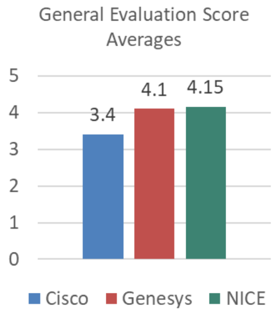
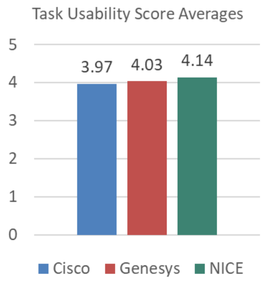
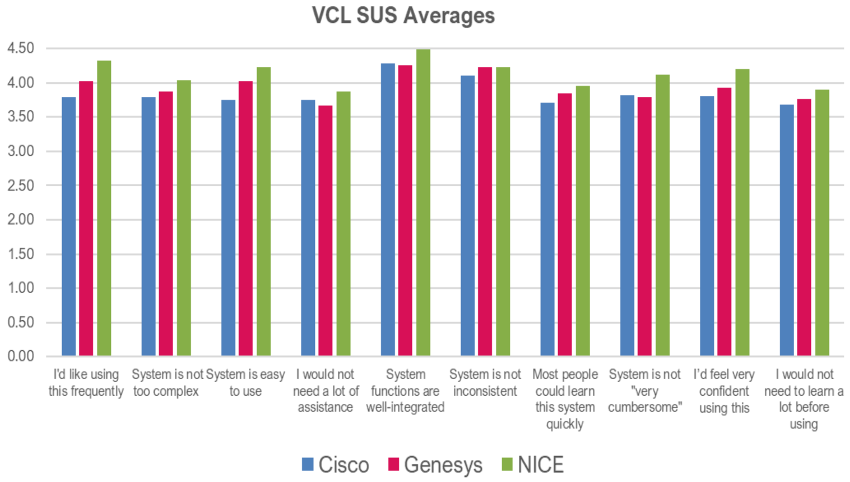
Veterans Crisis Line: Modernizing Mental Health Infrastructure

Results: Data-Driven Consensus

NICE and Genesys scored very similarly.







OIT felt Genesys would have more flexibility, would be easier to maintain / upgrade, and would overall work better for the future needs of the organization. The user scorecards gave valuable feedback on what training and change management the VA would need to implement in order palliate any user pain-points. Since the next phase of the project was change management, this was invaluable data to have.

Impact:

- VA OIT chose vendor based on technical + user evaluations
- Reduced technology selection timeline by 40% by eliminating rounds of stakeholder debate
- User participation increased buy-in for change management, allowing responders to become champions vs. resisters
- Established a replicable framework VA can now use for other contact center modernizations

What I Learned:

Navigating Bureaucracy Through Empowerment

Government procurement processes can exclude end users. By framing user research as "risk mitigation" and building evaluation tools that produced quantifiable data, we gave user needs a more prominent voice to a procurement team that typically only trusts technical specs.

Parallel Scorecards

Giving users and IT separate evaluation frameworks to calculate together and equally helped mitigate the "technical requirements override everything" dynamic.

Design Research as Change Management

The evaluation process was a change management intervention in itself. By making OIT alert teams to nominate “power users” for our study, management was now aware of the upcoming software change. Similarly, as I trained users on our evaluation methodology, I was simultaneously preparing them to be informed advocates for the future implementation.

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Mayo Clinic CRS Digital Monitoring

Design Problem:

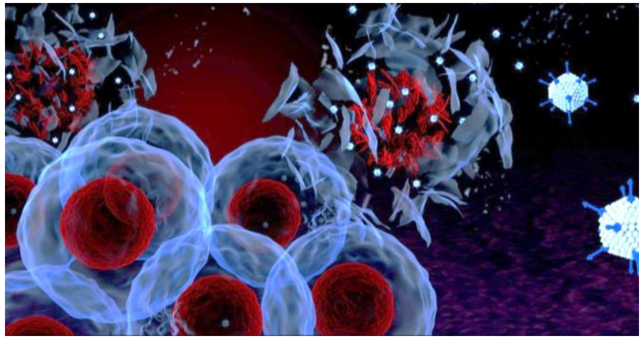
Cytokine Release Syndrome (CRS) is a common autoimmune response for a patient undergoing CAR-T cell Therapy or therapeutic antibody treatments for **Non-Hodgkin's Lymphoma**. The severity of CRS varies, with 4 stages defining the level of medical intervention needed (ranging from steroids to ER admission.) **Onset of CRS is typically within a week after treatment, so patient's vitals and health are monitored in a hospital or adjacent facility for 2-3 weeks.**

This creates significant **burdens** for:

Patient: Extended hospitalizations, travel requirements, quality of life
Healthcare system: Limited bed availability, high costs ($400K+/patient episode)
Clinical staff: Resource-intensive monitoring, staffing constraints
Accessibility: Geographic barriers limiting treatment availability

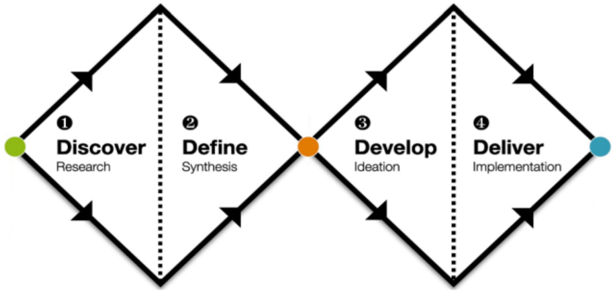
Design Challenge:

Create a prototype for a remote digital solution that enables the 2-week observation period to be done in the comfort of a patient's home, while maintaining clinical safety standards and improving patient outcomes.



Approach: Double Diamond Design Framework

Deep-diving into the problem to fully understood it before jumping to solutions, then iteratively developing and validating our approach.



Data utilized:

Primary issues identified:

My Role:

As Lead Design Strategist, I led a cross-functional team of 7 across 4 markets through research, service blueprint creation, and AI integration strategy within the care ecosystem. I facilitated 15+ stakeholder workshops, navigated international regulations, and translated clinical requirements into patient-facing experiences. I presented to executive leadership to secure pilot approval and developed the phased implementation roadmap with success metrics.

Solution:

Created and tested prototype that will reduce hospital stay for patients by up to 71%, both freeing up hospital resources for other billable treatments and uniting patients with their families sooner.

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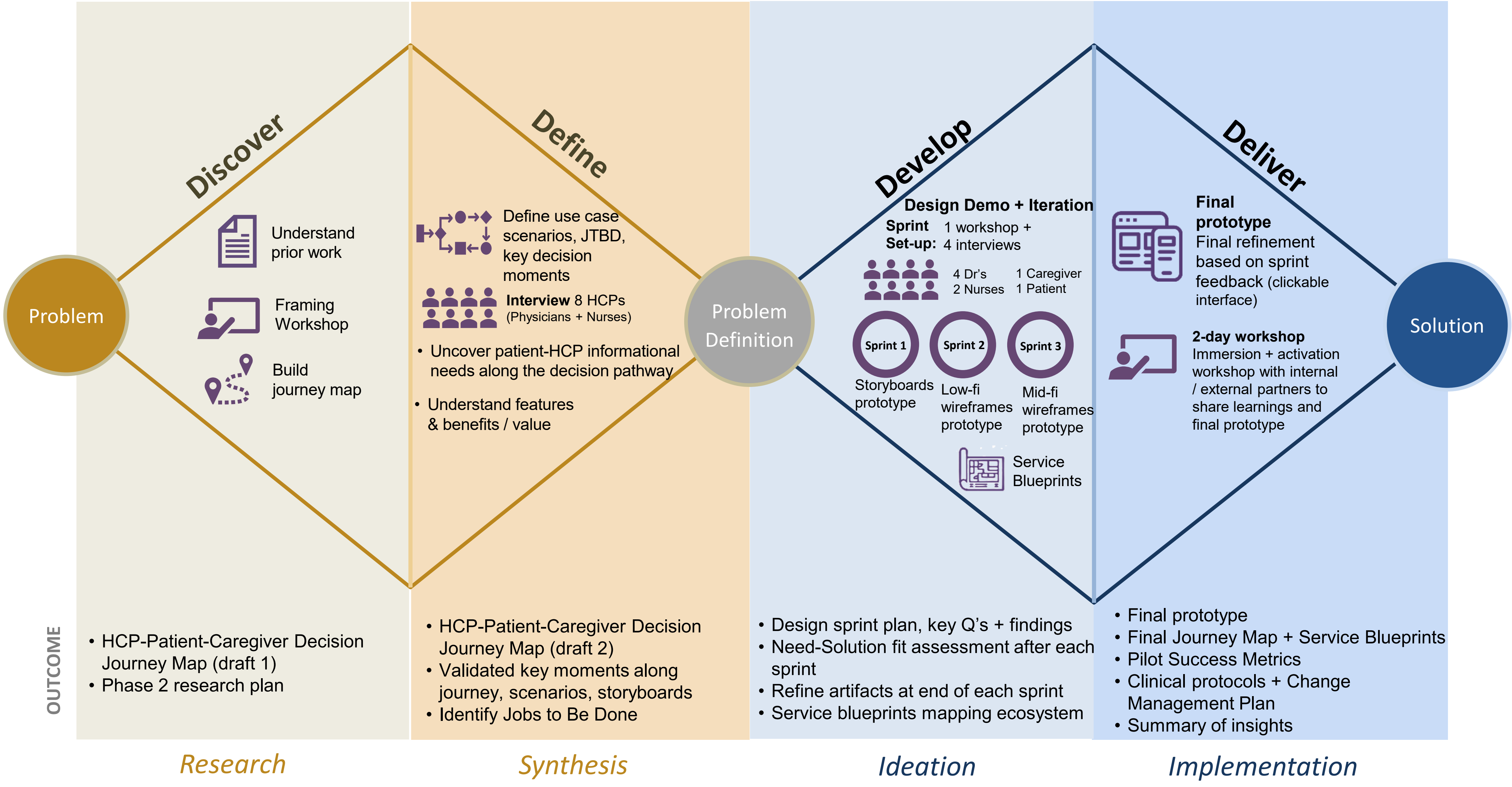
Mayo Clinic CRS Digital Monitoring

Approach: Double Diamond Design Framework



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Mayo Clinic CRS Digital Monitoring

Outcomes + Impact

Projected Clinical Impact

Patient Safety:
- Reduce time-to-intervention for complications through early detection
- Enable standardized monitoring protocols across all sites

Patient Experience:
- Reduce patient out-of-pocket costs with shorter hospitalizations
- Improve quality of life during recovery period
- Reduce caregiver burden by structured support + decision guidance

Hospital Operations:
- 71% reduction in required hospital bed days (from 21 → 6 days)
- Increase capacity for new CAR-T patients Enable treatment at community hospitals, not just academic medical centers

Health System Economics:
- Projected cost savings of $150K-$200K per patient episode
- Expand billable CAR-T treatment capacity without facility expansion
- Create new revenue streams through remote monitoring services

Business Value

Access Expansion: Enable treatment for patients previously excluded due to geographic barriers

Scalability: Create replicable model for deployment across Mayo Clinic network and partner institutions

Datasets: Build longitudinal dataset on CAR-T recovery patterns to inform future care improvements

Future Opportunities

Predictive analytics: Leverage longitudinal data to develop predictive models for complication risk

AI to personalize: Use machine learning to tailor monitoring intensity to individual patient risk profiles

Expansion to other treatments: Adapt model for other high-risk therapies requiring intensive monitoring

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Consular Affairs: Passport Agency : Transforming America's passport system + redesigning service delivery for 22M citizens

Design Problem:

For most Americans, applying for a passport is their only direct interaction with the federal government. It's a moment that should build trust, but instead, the experience often creates frustration. With **22 million passports issued annually and 6 new agency locations** planned nationwide, the U.S. Department of State's Consular Affairs Passport Agency (CA/PPT) faced a critical question: **How do we scale while improving both customer experience and employee satisfaction?**

Challenges:

- 29 existing passport agencies + 7,400 acceptance facilities (post offices, libraries)
- Multiple customer channels: urgent in-person, online renewal, mail-in, carrier
- 3 workforce types: federal employees, contractors, seasonal workers
- Congressional mandates affecting processing (gender policy holds, travel restrictions)
- Legacy technology causing frequent outages
- Fragmented processes and each agency operated slightly differently

My role: Project Co-Lead / Research Lead on a 15-person team, leading agency through tasks and conducting organizational transformation and service design for CA/PPT. I co-led project strategy and team member assignments, and led all qualitative research, site visit strategy, customer journey mapping, and workshop facilitation. I worked closely with LSS expert / co-Lead on all process mapping.

Data utilized:

- Congressional Mandates
- Surveys
- Interviews
- Focus groups
- Process Mapping
- Org Structure / Function Analysis

CA/PPT issues identified:

- Overlapping Functions
- Silos across Offices (esp HQ + HR)
- Resource Allocation & Planning
- Old Technology, Outages
- Inconsistent Training & develop.
- Staff Turnover

Customer issues identified:

- Transparency throughout process
- Electronic renewal UX
- Form submission clarity
- Urgent / Expediting Times
- WH orders delays (ex: gender, travel restrictions)

Goal:

Create a harmonized "future state" operating model that new agencies could adopt while improving coordination between headquarters and field offices, standardizing processes, and enhancing both customer and employee experiences. Conduct an independent assessment of human capital management, organizational structure, and customer delivery models. Ensure consideration of Online Passport Renewal (OPR) and its expansion to new eligible application types.

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Consular Affairs: Passport Agency : Planning an assessment around bureaucracy, ad hoc requests, and budget constraints

Approach: Building Consensus through Co-Creation

Given the complications of bureaucracy and silos in the government, our team knew the best way to get buy-in was to give both HQ and field offices ownership of the future state.

Data Analysis

- HQ + 6 Sample Sites
- Current Org Structure
- Policies
- Procedures
- SOP's

- Over 90 documents
- 12 current org structures

Interviews

- Over 120 Interviews with 150+ stakeholders
  - Issuance + Support
  - Philly, Charleston, DC, Hawaii, Tucson
  - SME's in LSS, CX, TR
  - 30+ focus groups

- Managing Directors
- Regional Directors
- HQ Directors
- Division Chiefs
- SME's

Sample Sites:
- Directors
- Assistant Directors
- Supervisors
- Managers
- Passport Specialists

Questionnaire

- 23 Division Chiefs surveyed (11 responses received)

Questions on:
- Org Structure
- 0967 Job Series
- Processes
- Customer Focus
- Workforce and Workload

Site Visit

- Philadelphia Nov 2024
- Charleston Feb 2025
- WN + SIA (D.C.). Mar 2025
- NPC (New Hampshire) Mar 2025
- Hawaii Apr 2025 *virtual*
- WPC (Tucson) Apr 2025 *virtual*

- Met with directors, assist. directors, managers, specialists
- Shadowed workers
- Toured entire process

PHASE 1

Initial Findings

- Assessed current baseline, applied observations and preliminary analysis

- As Is Current State Process maps + gaps
- Pain points
- Main themes

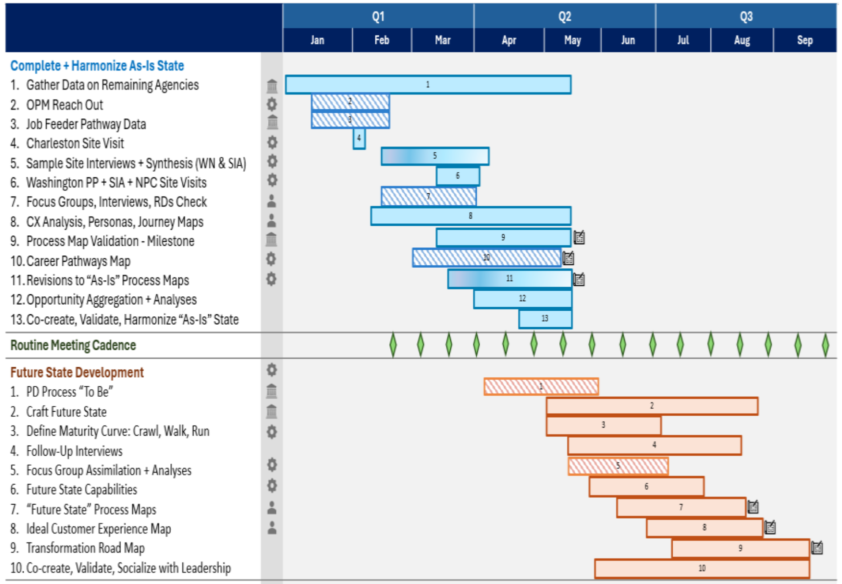
PHASE 2

Future State

- Co-creation sessions for Future State Map
- Buy-in with leadership Create Ideal Customer Journey Map
- Technology aspirations for future sites, online renewal roll-outs

- Future State Process map
- Ideal Customer Journey
- Online Renewal Map
- 6 new agencies org
- HQ/field comms protocol
- Org restructuring rec's

PHASE 3



PHASE 1: Immersion Across System (HQ + Field)

Research Design Decisions:
- Org structure breakdown: HQ functions + Field expectations
- 7 site visits to cover all CA/PPT specialties
- Shadowing at every level
- 120+ interviews with 150 stakeholders across HQ + Field
- Questionnaires, 30+ focus groups for honest feedback
- Weekly C-suite presentations for executive alignment
- Various mapping iterations

PHASE 2: Making Complexity Visible

Synthesis of 100+ hours of interviews, site observations, and doc analysis into visual artifacts to drive alignment:

A. Service Blueprints for 4 Customer Journeys:
1. New passport (adult, child)
2. Online renewal, if eligible
3. Urgent travel (expedited processing)

B. Process Map Consolidation
- 400+ steps down to 150

PHASE 3: Co-Designing the Future State

Over 15+ workshops were facilitated, where HQ and field staff co-created solutions:

Workshop Design:
- Present research findings without interpretation
- Ask, "What would need to be true for this to work better?"
- Capture competing priorities + negotiate trade-offs in real-time
- C-suite sponsors present to make decisions on the spot

This approach transformed resistance into ownership. Field offices weren't being told the future state, they were building it.

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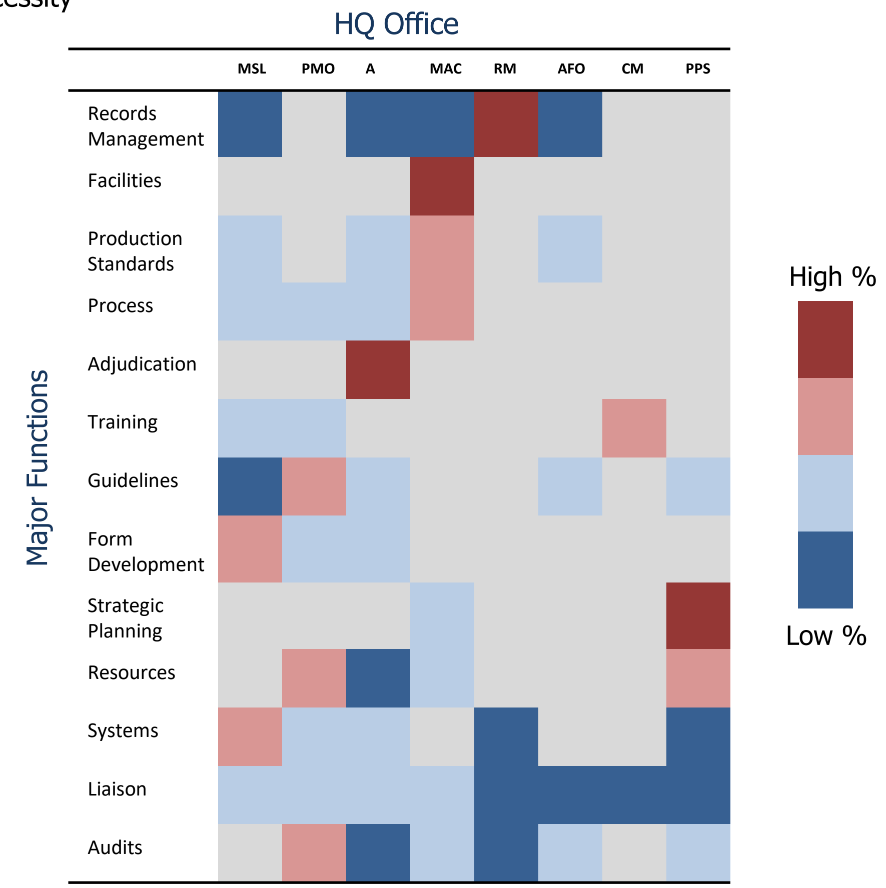
Consular Affairs: Passport Agency : (Mis)alignment between office purpose and functions

Conflict 1: Headquarter vs. Field Reality

Misalignment between office's purpose and functions can impact efficiency + priorities
- HQ thought liaison work was central to their role, but spent only ~15% of time
- Strategic planning lived in just 2 HQ offices, while 29 field offices expected ample guidance
- Training existed in parallel: HQ provided some, field offices developed own out of necessity

- All units rely heavily on HQ for appropriate guidelines in daily duties
- HQ believed they provided comprehensive strategic guidance
- Field offices felt abandoned, forced to create their own training programs and operational workarounds
- Functional analysis revealed disconnect between HQ and field offices on planning and resources



HQ Major Functions

Major Agency / Centers Processes

	Guidelines	Training	Adjudicate	Liaison	Production Standards	Records Mgmt	Form Dev.	Process	Audits	Systems	Resource & Strategic Planning
Book Printing/ Quality Check/ Mail Out/ Achieve (BQMA)	✓					✓				✓	
Case Adjudication (Adj)	✓	✓	✓	✓	✓				✓		
Customer Service (CSX)	✓			✓	✓		✓	✓			
Data Entry / Batching / Image (DBI)	✓						✓			✓	
DS-5055 Corrections (5055)	✓	✓	✓	✓	✓	✓					
Fraud Review (FR)	✓	✓						✓	✓		
Potentially Fraudulent Birth Docs (PFBD)	✓	✓	✓			✓					
Special Issuance for Military(SIM)	✓	✓		✓	✓						

Disconnected Functions

HQ View: Identifying redundancies to define roles and enhance communication
Leadership identified tasks, grouped by function + identified as a % of each function across HQ offices

HQ & Field : Streamlining Operation and Enhancing Communication
Fragmented coordination and unclear role delineation between HQ and Field Offices hinders consistent execution

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Conflict 2: Standardization vs. Specialization

Lessons from the field: real user behaviors, operational limits, and site priorities based on specialty:

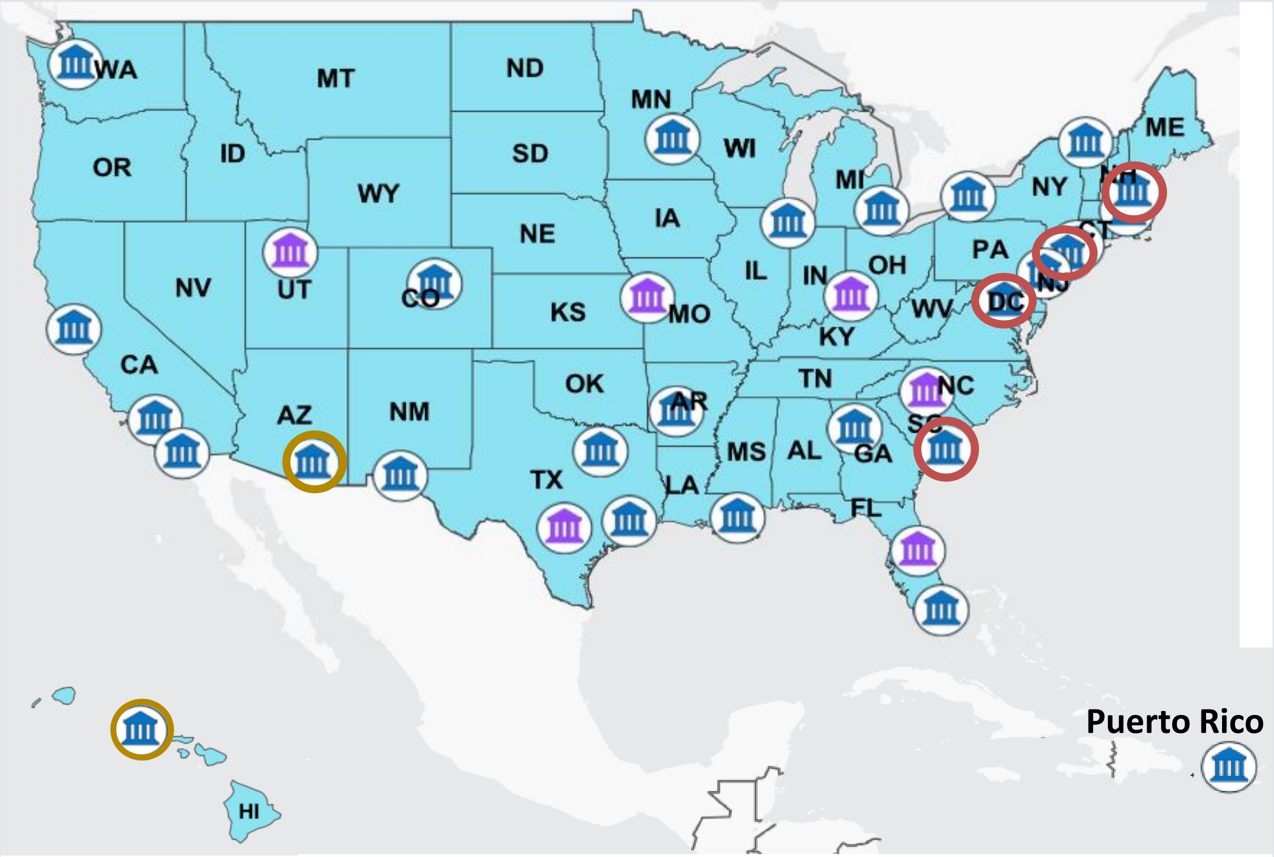
Touring Sites to Understand Their Specialties:

- 7 / 29 Passport Agencies sites were toured
  - Tour sites chosen to cover all specialties + budget / sponsor constraints
  - Western virtually toured because of budget / time / notice
- Tours shadowed day-to-day processes, group interviews + widget process map analysis
  - 1-on-1 Leadership interviews in advance of each tour

Toured:	Specialties:
Philadelphia	Amish / Mennonite Applications
CPC (Charleston) *Mega Print Center*	Potentially Fraudulent Birth Certificates (PFBD), 5504 (Corrections), Special Issuance for Military
Washington D.C.	Expedited Counter, "VIP" applicants (congress, senate, etc.)
Special Issuance Agency (SIA)	Diplomatic, Official, Service, Military no-fee, Military dependents, Emergency 1-yr
NPC (New Hampshire)	Military no-fee, airline crews
Virtual	**Specialties:**
Hawaii	Adjudicating nationals
WPC (Tucson) *Mega Print Center*	Repatriation

Challenge: How do you create a "single source of truth" process map when legitimate operational differences exist?

Geographical Locations:



Established location

Future location

- Salt Lake City, UT
- Kansas City, MO
- Orlando, FL
- Charlotte, NC
- San Antonio, TX
- Cincinnati, OH

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Consular Affairs: Passport Agency : Interview + focus group data

Our qualitative research uncovered some key takeaways:

Interviews Summary:

	HQ	Field	Sample Sites	Takeaways
Communications	• HQ and Field silos • Transformation priorities vs. daily operations comms	• Transformation comms do not consistently reach employees • Silos between IT and Ops	• Insufficient comm from HQ about initiatives and anticipated workloads • Day/night shift comm and equity gaps	• Enhance coordination and consistent communication between HQ and Field • Align transformation initiatives and daily operations effectively
Transformation	• Transformation priorities are misaligned with daily operations • Lack of long-term vision to sustaining large projects	• Field's input on transformation is not always gathered ahead of implementation	• Transformation projects not aligned with some site priorities • Focus has been on specialist needs, overlooking management needs	• Greater alignment of transformation projects with organizational priorities • Focus on project sustainability to ensure long-term success
Roles & Responsibilities	• Duplication of efforts and unclear duties for shared functions • Insufficient support staff and FTE's to meet org needs • Inaccurate PD's, unsustainable 0967 job family • High contractor turnover	• Lack of consistent standards across agencies • Resources allocated only to those who request it • Ongoing issues with Union	• Lack of locality pay creates challenges in recruitment / retention • High staff turnover, esp. among D.C. staff (people use as stepping stone to receive security clearance)	• Clarifying roles, responsibilities, and Position Descriptions will help to address duplication of efforts and better utilize talent at all levels
Process & Technology	• Outdated systems and tools reduce operational efficiency	• Current platforms outdated / unstable • Workload management challenges persist, requiring skilled analysts • Non-standardized resource distribution • OPR should consider employee experience	• Technology outdated - "we've grown into a culture of workarounds" • TDIS outages • Need better process for triaging, tech holds, and undeliverable mail	• Streamlining op processes, like case triaging and hiring, to reduce delays and improve efficiency • Modernize outdated systems • Invest in stable and scalable platforms to support current / future demands
Skills & Learning	• Insufficient tech knowledge to support transition to Agile • Want more training opportunities for specialists beyond mandatory and off-peak • Need better way to track training	• Supervisors need adequate training, resources, and SOP's • 'Discretion' for adjudication is inconsistent across agencies	• Removal of individual production standards affecting morale • PFBD needs more targeted training, more support • Training for supervisors	• Expand targeted training opportunities and address both specialists and management needs to enhance workforce capability and support organizational growth

Focus Group Data Summary:

Organizational Structure

- Resource inefficiencies (technology skills, workforce planning)
- Focus is on transformation changes, not on improvement opportunities
- Need a prioritized roadmap

Engagement

- Field agencies have training programs to make up for training HQ does not provide
- Supervisors do not feel equipped with appropriate training to lead / guide employees when promoted into their roles

Communications

- Siloed communications between HQ + Field
- Transformation efforts fluctuate based on employee chain of command
- Absence of formalized path for Field agencies to communication + cross collaborate

Workforce + Workload

- Career ladder promotions can be fast based on increased vacancies for passport specialists
- Position Descriptions (PD's) not indicative of all work being performed because "other duties as assigned" has wide range of responsibilities

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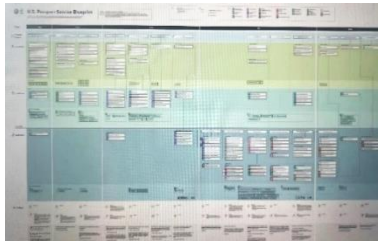
Consular Affairs: Passport Agency : Mapping CX , widget process, + EX to expose pain points and decision moments

Conflict 3: Customer Trust vs. Bureaucratic Opacity

CA/PPT's wants Passport experience to **earn citizens' trust. Customer's goal is simply unencumbered travel**. Blueprints and journey maps were created to help drive alignment.

Customer Journey Maps

New Passport (Adult)



New Passport (Child)

Passport Renewal (Online eligible)

Passport for Urgent Travel

Each blueprint mapped:

- Customer actions and pain points, front office employee touchpoints, back office processes, supporting systems and technology

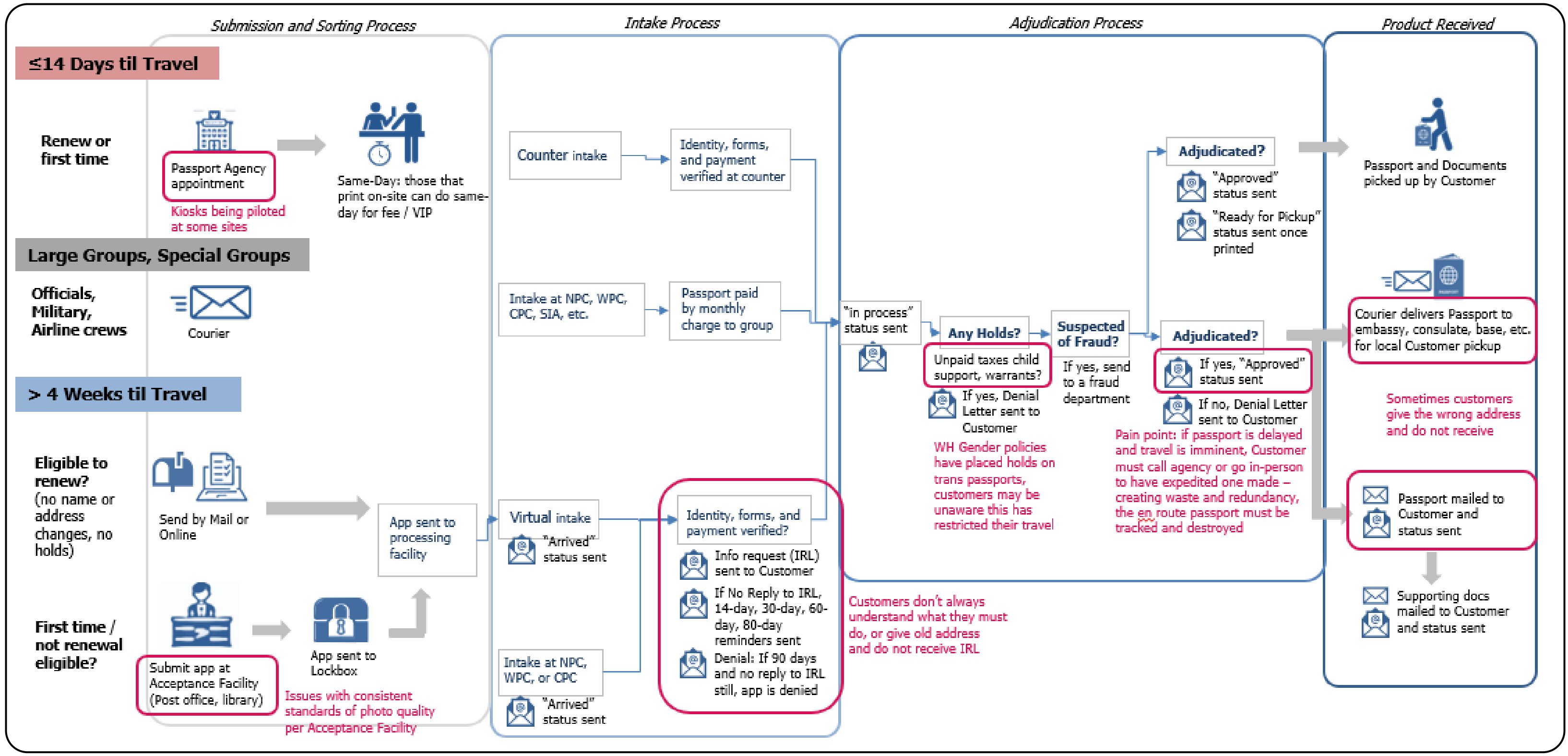
Based on data, we knew Customers' expectations:

- Transparent status updates + customer action needed at every stage
- Mobile-friendly online applications, easy from laptop or phone
- Quick resolution for urgent travel needs

Visual artifacts revealed Customers' true experience:

- Unclear form requirements
- "Black box" processing with minimal status updates
- Policy holds applied without customer notification
- Address changes causing missed communications and denials
- Inconsistent customer experience across touchpoints
- CX survey metrics inconsistent, may fail to find root causes of dissatisfaction

High-level Widget Journey + Customer Touchpoints



Employee experience directly impacts CX and PPT operations. **Interviews and site visits revealed employee pain points:**

- Technology outages
- Communication issues with HQ
- Inaccurate Workload prediction models (impacting training and flexibility)
- Seasonal variations (peak is May to Sept.) affect workload, employee schedules + passport turnaround

Key Insight: The system was designed around internal processes, <u>not</u> customer needs

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Consular Affairs: Passport Agency : Process map consolidation

Process Analysis: End-to-End Process Map Revisions - Using employee experiences to inform future harmonization

Lean Six Sigma best practices, site walk-throughs, and SME workshop validations shaped our initial end-to-end map.

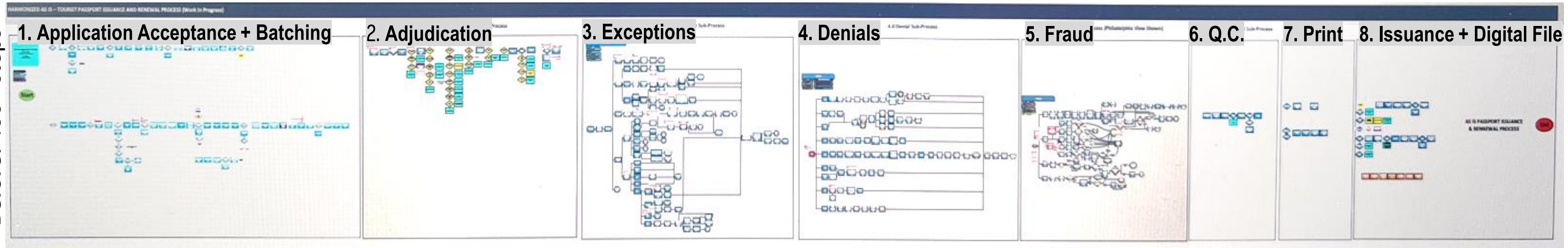
Since each of the 29 agencies had a slightly different process, it was imperative that the new 6 agencies had a concise, harmonized map (that still accommodated specialties) to build their processes around.

Process Map Consolidation:

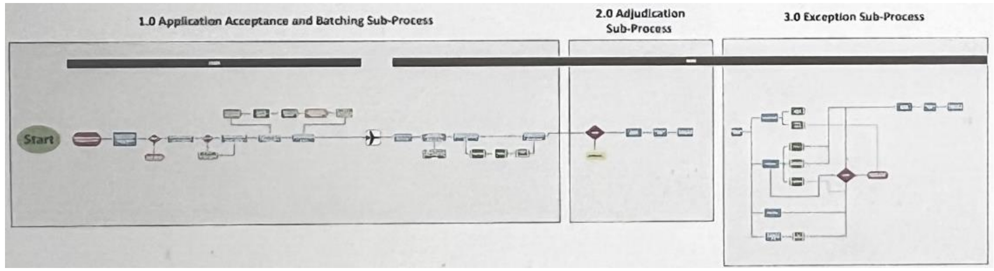
Working with our Lean Six Sigma lead, I facilitated workshops to streamline the 400+ documented process steps down to ~150, across 8 core subprocesses:

1. Application acceptance + batching
2. Adjudication
3. Exceptions handling
4. Denials
5. Fraud review
6. Quality control
7. Printing
8. Issuance + digital filing

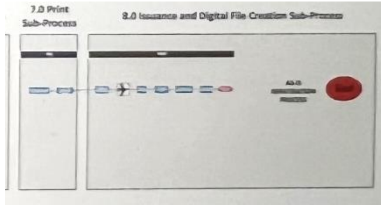
Before: 400+ steps



After: Streamlined



Contribution to this project concluded at this stage



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IMMEDIATE IMPACT:

Harmonized Operating Model
- Reduced process complexity from 400+to 150 steps while keeping specialty flexibility
- Created "single source of truth" process maps for 6 new agencies
- Established clear role delineation between HQ strategic functions + field op execution

Digital Strategy for Scale
- Informed OPR’s expansion to new eligible application types
- Recommended transparent status updates at each stage to build citizen trust

Organizational Alignment
- Research findings used to justify HQ restructuring proposal
- Weekly C-suite presentations maintained executive buy-in throughout project
- 30+ stakeholder groups aligned on implementation roadmap and sequencing

PROJECTED IMPACT:

- 14% increase in online application adoption (from 8% to 26%), reducing acceptance facility burden and accelerating processing
- 6 new agencies opening with standardized processes (vs. historical "figure it out" approach)
- Improved employee satisfaction through clearer HQ guidance and resource allocation

“This map gave us a shared language to talk about solutions. For the first time, HQ and Field were looking at the same picture."
- CA/PPT Assistant Director

What I Learned:

Balancing Standardization with Operational Reality
Field offices taught me that standardization without flexibility just creates more workarounds. The "harmonized" process map we created honored necessary variation while eliminating arbitrary differences.

The Power of Visual Artifacts in Bureaucracy
In government, change requires extensive stakeholder alignment. I learned that a single well-designed diagram can accelerate months of debate by making invisible problems visible.

Co-Creation as Change Management
Workshops with CA/PPT SME’s and management was an opportunity to start the change management process. By having field offices co-design the future state, we were creating buy-in and ownership from the start.

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Rethinking Hospital Waits: Designing a Patient-first Arrival Experience at Northwell Health

EDRA CORE 2022 Award

Design Problem: Using Simulation + Service Design to Reinvent Outpatient Care

In the traditional outpatient hospital model, each floor and department operates independently with its own registration desk and waiting area. A patient attending two specialist appointments on the same floor experiences the process twice: lobby wait, elevator queue, floor registration, waiting room. This could be a half hour of non-clinical time before even seeing the first provider, then the cycle repeats. This fragmented model creates systemic inefficiencies, such as congested floors, duplicated staff roles, underutilized space, and poor patient experience.

Design Challenge: Northwell Health, one of New York's largest health systems, is renovating a 260,000 ft² multispecialty office building in Manhattan to serve its urban patient population. Leadership has an ambitious vision: **embrace emerging technology, rethink the traditional hospital arrival experience,** and create spaces that **encourage staff collaboration and flexibility**. Our team wanted to explore all possibilities and ask: **What if we challenged the fundamental assumption that every floor needs its own waiting room?**

Data gathered:

- Population Data
- Computer Simulations
- Best Practice Research

Primary issues identified:

- Traffic Flow
- Seating
- Department Needs
- Wait Time

Design challenges:

- Constraints of an existing building
- Elevator capacity
- Centralized vs. De-centralized

Goal: Design an innovative space program for a 260,000 ft² building that:

- Reduces patient wait times to under 2 minutes at check-in
- Improves patient flow and wayfinding
- Optimizes elevator capacity and lobby congestion
- Frees up clinical floor space for actual care
- Maintains check-in staff utilization at or below 85% (preventing burnout)
- Validates design decisions with data before construction

My Role: Senior Healthcare Design Researcher at EwingCole (an architecture firm specializing in hospitals and cancer centers). Over 6 weeks, I led research and computational modeling to test whether a centralized registration and waiting model could improve patient flow, reduce wait times, and create a better experience without creating new bottlenecks.

Solution:

Based on department interviews, stakeholder meetings, journey mapping, and scenario and simulation modeling, an innovative program was developed. **Floors were sorted into clinical spaces and academic villages, with a main centralized registration** on the 1st and 2nd floors to control traffic volumes while welcoming patients into the health system.

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Rethinking Hospital Waits: Northwell Health Building Program

EDRA CORE 2022 Award

Design Challenge: When Convention Creates Chaos

Through research, journey mapping, and site observations, I identified how the traditional outpatient model created cascading problems:

PROBLEM 1: Fragmented Patient Journeys

Traditional Model (Every floor has Registration + Waiting):

- Patient sees primary care on Floor 3, then cardiology on Floor 5, then lab work on Floor 2

Each stop requires: Finding the department → Checking in again (even though pre-registered) → Sitting in a new waiting room → Navigating elevator congestion between floors

Impact on UX: Patients spends too much time trying to navigate building, each floor compounding the issue.

PROBLEM 2: Elevator Capacity Frustrations

With patients traveling between floors for every appointment:

- Morning rush (8-10 AM) creates 8-12 minute wait times for elevators
- Patients can run late to appointments if they can't get upstairs

Impact on UX: The building's vertical circulation becomes a bottleneck that affects the entire patient experience.

PROBLEM 3: Inefficient Use of Clinical Space

When every floor dedicates 30-40% of space to registration desks and waiting areas:

- Less room for exam rooms, treatment spaces, and provider offices
- Duplicated staff (registration clerk on every floor)
- Underutilized waiting areas during off-peak hours
- Congested waiting areas during peak hours with no flexibility

Impact on Design: Clinical space is sacrificed for administrative functions, meaning the building design is serving the process more than the patient.

PROBLEM 4: Existing Building Constraints

Since Northwell was renovating an existing structure, we needed to design around:

- Fixed elevator shafts, so we couldn't add more capacity
- Immovable structural columns, constraining floor layouts
- Limited HVAC and mechanical flexibility
- Height restrictions

Impact on Design: With fixed infrastructure, it was all about how we could design *smarter,* not bigger.

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Rethinking Hospital Waits: Northwell Health Building Program

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Hypothesis: What if we centralized all registration + waiting on Floors 1-2 and dedicated Floors 3+ entirely to clinical care?

Potential Benefits:
+ One check-in for all appointments → better patient experience
+ More clinical space on upper floors → better care delivery
+ Flexible waiting capacity → scale up/down based on demand
+ Reduced elevator trips → patients go straight to clinical floors

Potential Risks:
- Lobby overcrowding → all patients in one place
- Longer elevator waits if everyone goes up simultaneously
- Check-in bottlenecks → check-in kiosks in lobby full during peak?
- Wayfinding confusion → patients unfamiliar with centralized model

Process: Designing with Data to De-Risk Innovation

PHASE 1:

Current State Research
- Site observations
- Best practice research of hospitals experimenting with centralized models

PHASE 2:

Journey Mapping
Mapping current patient experience and pain points with registration, wait times, and building wayfinding

PHASE 3:

Computer Simulation Testing
Testing of traditional and centralized models in FlemSim software

PHASE 4:

Present to Client + Refine
Pitch findings to C-suite to gather feedback on design direction to refine design concept

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Rethinking Hospital Waits: Northwell Health Building Program

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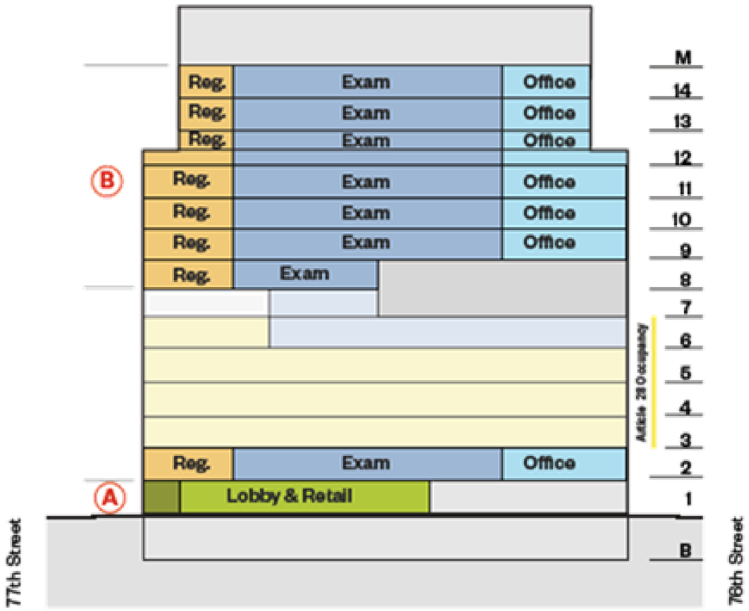
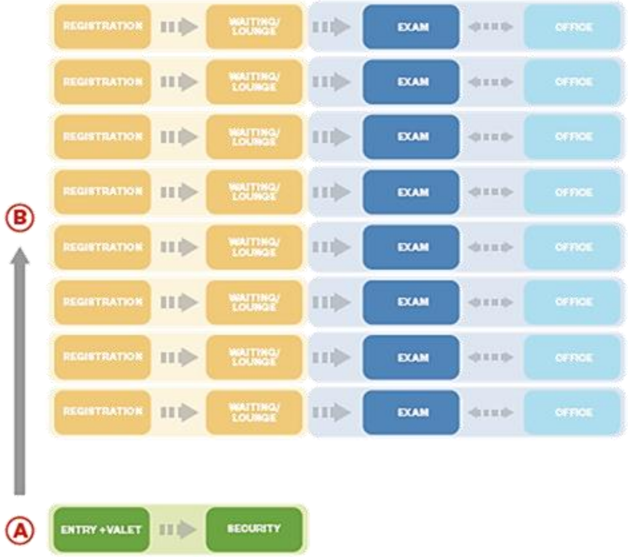
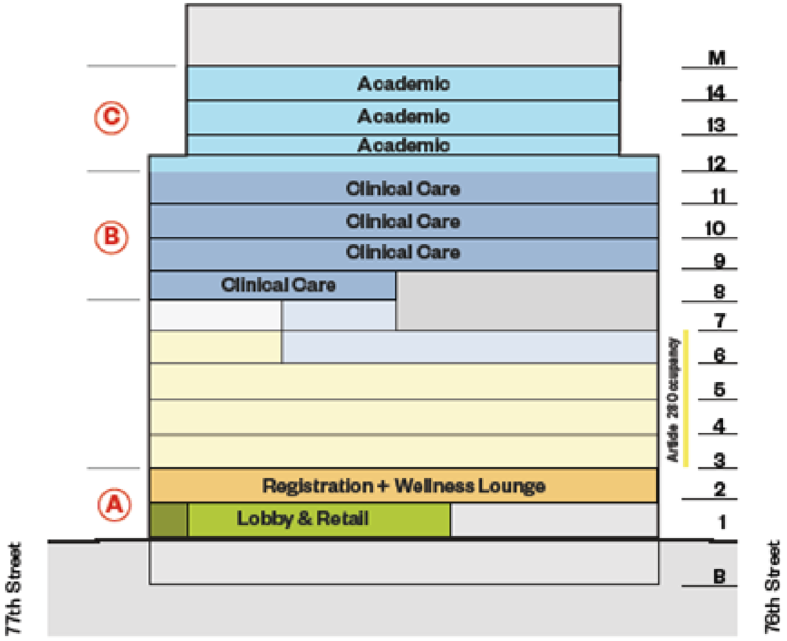
1. Model Redesign: What if we re-imagined the patient experience by centralizing registration and waiting? How could that improve patient flow and the patient experience?

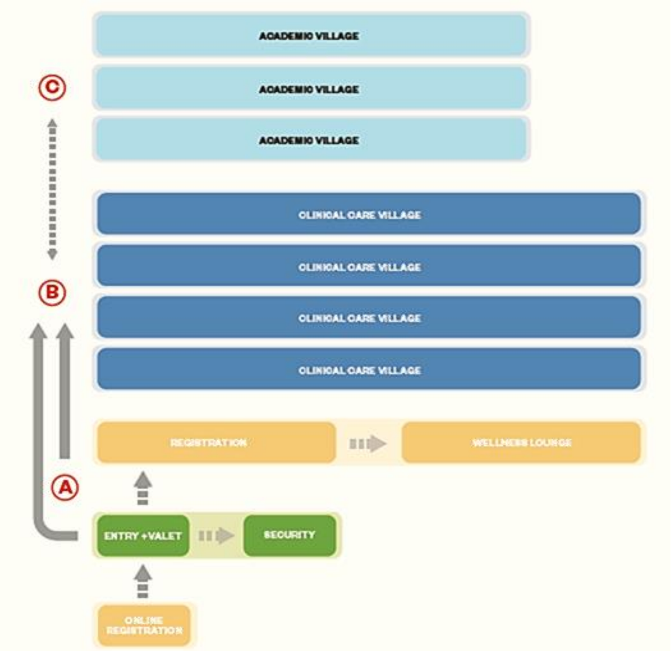
Before: Rethinking the conventional outpatient model, where a typical hospital has registration and waiting lounges on every floor, often for each department





After: With a centralized lobby and offices moved to separate floors, clinical spaces can be dedicated to patient care and experience reduced congestion





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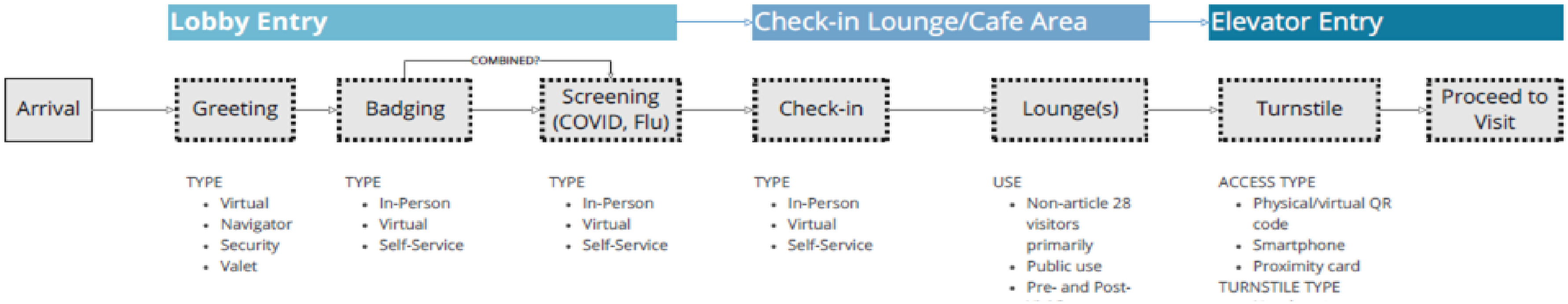
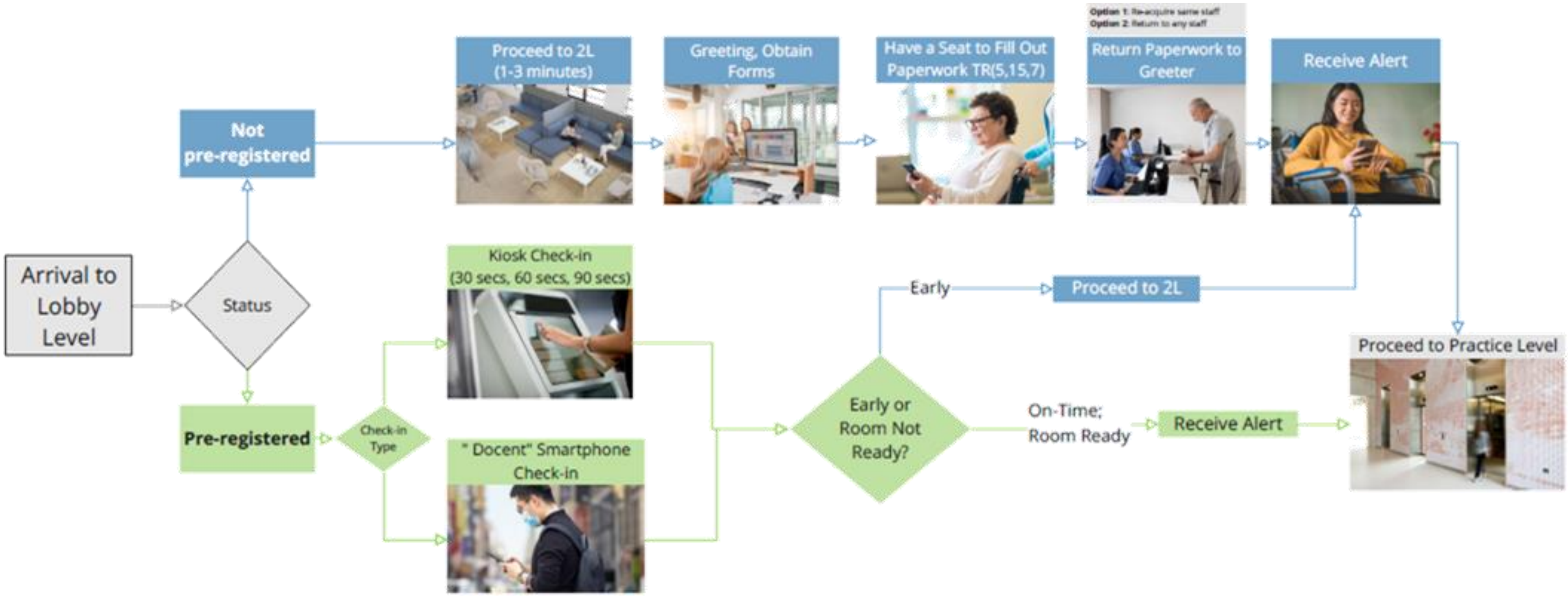
2. Journey Mapping: The patient experience was simulated using a journey map to explore where pre-registration could potentially expedite check-in, minimize wait times, and improve patient flow. **What could the check-in experience ultimately look like?**





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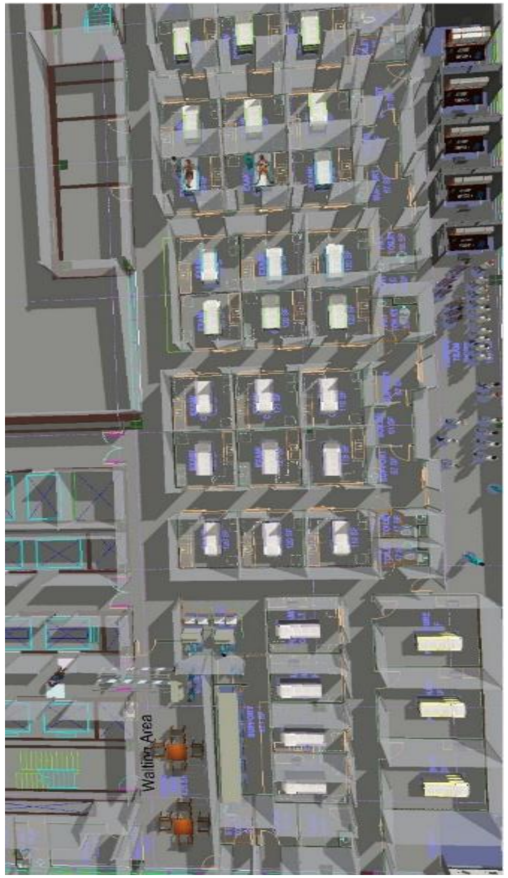
Rethinking Hospital Waits: Northwell Health Building Program

3a. Scenario and Simulation Modeling:

In Flexsim (3D simulation software), we created mockups of waiting areas and hospital floors, imported patient visit data for each department and pre-registration rate assumptions, and simulated an average week in the hospital.

Using pre-registration assumptions, we tested 3 scenarios to examine estimated bottlenecks, traffic patterns, and utilization rates. **This helped us identify peak times and maximum seating demand on any given day.**



Scenarios Tested:

- **Baseline:** 30% pre-registered
- **Medium:** 40% pre-registered
- **High:** 50% pre-registered

Objectives:

- Check-in wait time < 2 mins
- Check-in staff utilization ≤ 85%

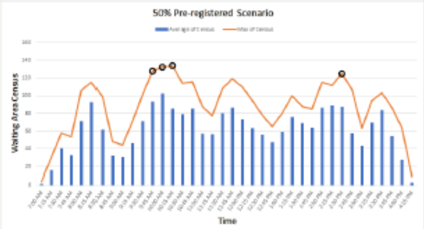
Comparing 3 pre-registration scenarios:

Pre-registration rate

50%

Seat demand during 4 highest peaks:
Range = 115 -134
Average = 125

- Waiting Areas
  - 8 -9AM Peak = 115 (Avg Peak= 93)
  - 9-10AM Peak = 128 (Avg Peak=93)
  - 10-11AM Peak = 134 (Avg Peak= 102)
  - 2-3PM Peak = 125 (Avg Peak = 89)



40%

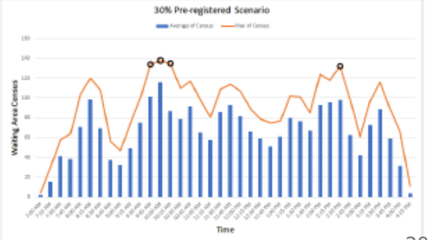
Seat demand during 4 highest peaks:
Range = 128 - 134
Average = 131

- Waiting Areas
  - 9 -10 AM Peak = 128 (Avg Peak= 96)
  - 10 - 11 AM Peak = 132 (Avg Peak=104)
  - 11- 12 AM Peak = 132 (Avg Peak= 95)
  - 12 - 1 PM Peak = 134 (Avg Peak = 89)

30%

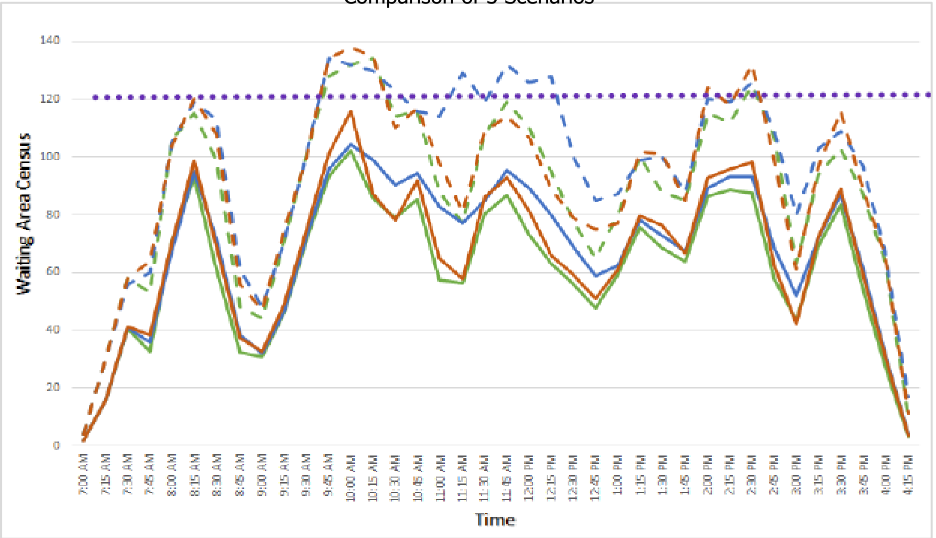
Seat demand during 4 highest peaks:
Range = 120 - 138
Average = 131

- Waiting Areas
  - 8 - 9 AM Peak = 120 (Avg Peak= 99)
  - 9 - 10 AM Peak = 134 (Avg Peak=101)
  - 10- 11 AM Peak = 138 (Avg Peak= 116)
  - 2 - 3 PM Peak = 132 (Avg Peak = 96)



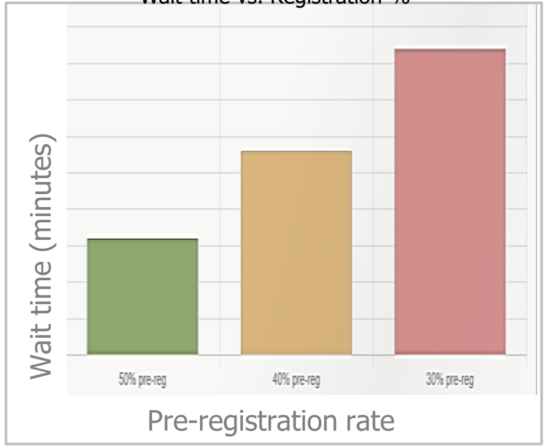
n=2045

Comparison of 3 Scenarios



- 50% pre-reg census average
- 50% pre-reg census max
- 40% pre-reg census average
- 40% pre-reg census max
- 30% pre-reg census average
- 30% pre-registered census max

Wait time vs. Registration %



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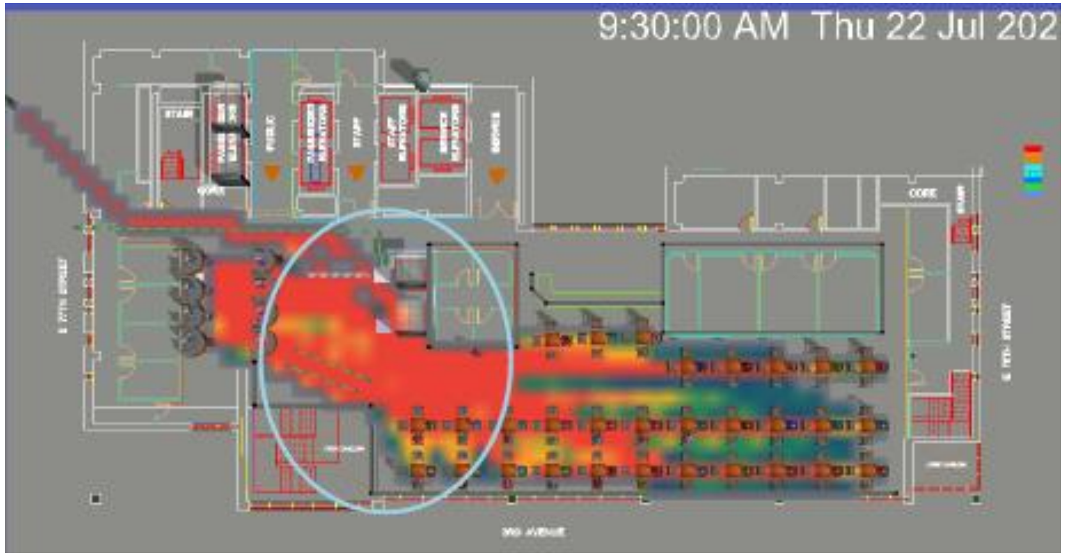
Rethinking Hospital Waits: Northwell Health Building Program

3b. Scenario Modeling: congestion and circulation



Key Findings from Simulation:

1. Pre-Registration is Essential for Centralized Model

X 30% pre-registered: Check-in wait = 4.2 min avg
X 40% pre-registered: Check-in wait = 2.8 min avg
✓ 50% pre-registered: Check-in wait = 1.6 min avg

Centralized model only meets <2 min check-in wait if Northwell invests in digital pre-registration tools (mobile app, patient portal).

Recommendation: Prioritize pre-registration UX in parallel with building design.

2. Flexible Seating Prevents Overcrowding

Heat Map Analysis showed:

- Average census: 45-60 patients in lobby at any time
- Peak census (9:15 AM on Mondays): 95-110 patients
- Traditional formula (seats = 1.5x avg census) would provide 90 seats but not enough for peaks

Recommendation: 120 seats with flexible furniture that can be reconfigured for other events during off-peak hours.

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Rethinking Hospital Waits: Northwell Health Building Program

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Key Findings from Simulation:

3. Elevators Can Handle Traffic (Barely)

Simulation showed:

- Elevator wait times increased from 2.3 min (traditional) to 3.8 min (centralized)
- Still acceptable but required staggered appointment scheduling to prevent 9 AM rushes

Recommendation: Implement 15-minute appointment buffer zones, such as 8:00, 8:15, and 8:30 start times.

4. Staff Utilization Stays in Safe Zone

With centralized model:

- Check-in staff utilization: 78% (below 85% burnout threshold)
- Traditional model required 8 staff across floors
- Centralized model needs only 5 staff (3 FTE cost savings)

Recommendation: Redeploy saved staff to patient navigation roles, such as helping patients find departments.

Results

Design Recommendations Adopted:

1 **Centralized Registration Model**
- Floors 1-2: Unified lobby with 5 registration desks
- Floors 3+: 100% clinical space (no waiting rooms, no registration)
- Flexible 120-seat waiting area with mixed use potential

2 **Digital Pre-Registration Strategy**
- Kiosks in lobby for walk-in patients
- Mobile app development prioritized

3 **Staggered Appointment Scheduling**
- Appointment block optimization by department
- 15-minute buffer zones to prevent elevator rush

4 **Wayfinding and Navigation Enhancements**
- Digital signage showing appointment locations
- Color-coded floors (clinical vs. administrative)
- App to help with wayfinding
- Patient navigators stationed in lobby (redeployed registration staff)

5 **Clinical Floor Space Optimization**
- Staff collaboration "academic villages" on upper floors
- Infusion center: 41 chairs (based on sensitivity analysis)

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Rethinking Hospital Waits: Northwell Health Building Program

EDRA CORE 2022 Award

Recognition: EDRA CORE Award 2022:
Environmental Design Research Association
Recognized for innovative use of computational modeling and evidence-based design in healthcare facility planning

What I Learned:

Data as a Design Tool, Not Just Validation
Simulation was initially chosen as a way to validate our design ideas. After running scenarios, however, we found it be a great generative design tool. The data / feedback allowed us to surface new solutions, such as the 50% pre-registration threshold and staggered scheduling.

Challenging Industry Norms Requires Proof
Design failures affect patient care, so we could not just tell architects "let's try something new" without evidence. Showing them FlexSim heat maps of lobby congestion, staff utilization charts, and wait time graphs gave the team confidence to explore other solutions.

Service Design Extends Beyond the Building
The centralized model only works if Northwell invests in pre-registration UX. This emphasized the interconnection between physical space design and digital service design, so I learned to make these dependencies explicit in my recommendations.

What I'd Do Differently:

Prototype the Wayfinding Strategy
We recommended digital signage and patient navigators but did not prototype how patients would actually navigate from lobby to clinical floors. A low-fidelity walkthrough with test participants could validate /challenge any wayfinding assumptions.

Test More Extreme Scenarios
We modeled typical weeks, but hospitals have surge events, such as flu season, community health fairs, and other emergency influxes. Stress-testing the centralized model under extreme volume would have surfaced potential failure modes.

Quantify ROI More Explicitly
We identified 3 FTE savings and 25% more clinical space, but not an explicit dollar amount or patient volume capacity increase. Expressing our findings in revenue potential and cost savings would help strengthen the case for our innovative design.

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Veterans Affairs: VA Profile : Building trust through reliable data, redesigning how 16M vets access their benefits

Design Problem:

When a veteran updates their address, that single change must update across 40+ VA systems, including healthcare appointments, medication delivery, disability payments, education benefits, and home loans. If data doesn't sync, a veteran might miss a critical doctor's appointment or not receive their monthly check.

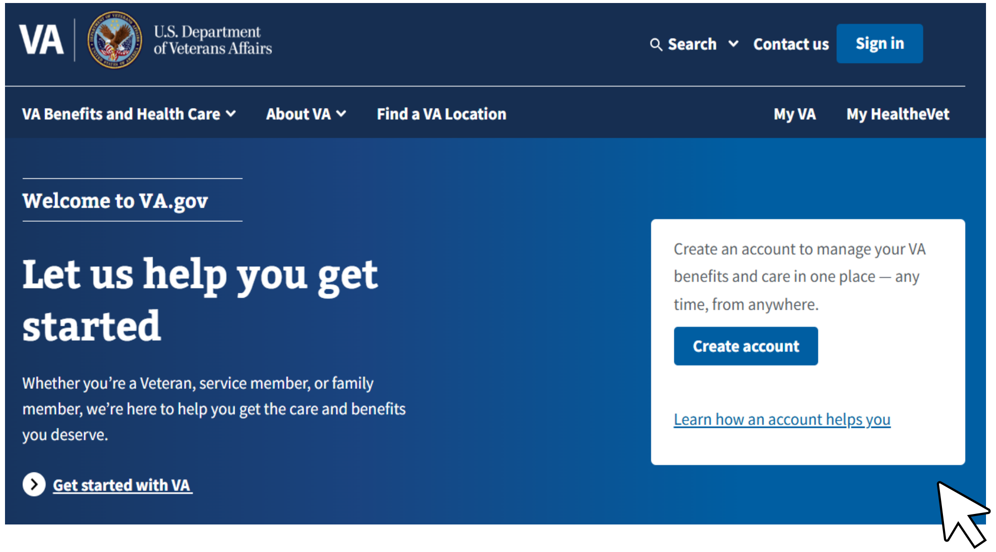
VA Profile is the authoritative source managing contact information for 16 million veterans, with plans to add millions more dependents and caregivers. The infrastructure is invisible, until it breaks and veterans lose access to life-sustaining services.

The Veterans Experience Office (VEO) set a new goal: earn 90% customer trust in VA Profile. Underlying issues still needed to be addressed to achieve this goal:

- Data conflicts across programs
- Stakeholder buy-in and approval
- Increasing awareness and adoption among internal customers
- Unclear roles and responsibilities for integration
- Development, testing, and production deployment

With such ambitious expansion plans, VA Profile needed to transform from a small custom operation into a scalable, reliable platform.

My role: Service Designer on a 4-person team conducting organizational assessment and process redesign for VA Profile's intake and integration operations. Over 3 months, I led stakeholder interviews, process mapping, and recommendation development to create a repeatable playbook for partner onboarding.



Goal:

Design a robust intake process that enables VA Profile to scale from 40 to 100+ partners while maintaining data integrity, increasing stakeholder adoption, and building the 90% trust threshold for veterans.

Data utilized:

- Best Practice Research
- Interviews
- Workshops
- RACI + Org Roles
- Process Mapping

Primary factors addressed:

- Business / IT Process Integration
- Data Management
- Partner Experience
- Knowledge Management
- Change Management

Solution:

36 recommendations were developed and prioritized through a combination of assessing potential impact, level of effort to implement, and timeframe to execute. The recommendations and prioritizations were validated through a human-centric design (HCD) Importance-Difficulty exercise with representatives from key roles across VEO.

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Veterans Affairs: VA Profile

Assessment Framework:

ACTIVITIES

Discover

Scope

- Clarify problem + objectives
- Identify key teams, user groups, and stakeholders
- Recognize areas of opportunity

Define

Data Gathering

- Review 31 documents + artifacts
- Interview 27 functional SMEs for pain points
- Conduct research on user-centric best practices

Develop

Synthesize & Review

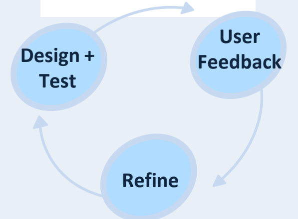
- Develop current state process maps + RACI
- Analyze interview data to map processes + interactions
- Facilitate stakeholder sessions to prioritize + validate

Deliver

Recommend

- Create and prioritize actionable recommendations by impact and required effort
- Identify + align best practices + expert insights
- Build playbook outline

Implement



- Approve and implement selected rec's
- Draft Playbook based on initial outline
- RACI review + update with stakeholders
- Design, test and implement targeted improvement(s) using rapid customer feedback

Critical Insights from Discovery:

Unclear, Non-Uniform Process

VA Profile has 6-phase integration process:

1. Awareness - stakeholder communication
2. Business Intake - requirements gathering
3. Data Intake - validate contact data
4. Development, Testing, Fielding
5. Technology Intake - configure solutions
6. Continuous Sustainment - test user stories

Interviews across teams revealed:

- Intake happened a variety of ways
- No clear owners for each phase
- Phases overlapped rather than sequential

Data Validation Not Systematic

Tracing 11 recent integration examples revealed:

- 36% had data quality issues in first 3 months
- No standardized protocols before deployment
- Manual checks instead of automated validation
- No feedback loops to partners when data conflicts occurred

Missing RACI

There was no uniform answer to "Who's responsible for stakeholder communication during integration?" There was no RACI (Responsible, Accountable, Consulted, Informed) matrix defining the roles.

Tasks fell through cracks because partners didn't know who to contact and departments didn't know it was their turn. Integrations were often delayed weeks or even months as they waited for approval.

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Veterans Affairs: VA Profile

Design Challenge – Scaling an Ad-Hoc System

VA Profile currently operated under individual expertise and workarounds, but this wasn't a scalable approach. Through discovery, we found three critical failure modes:

1) Every Integration is a Snowflake

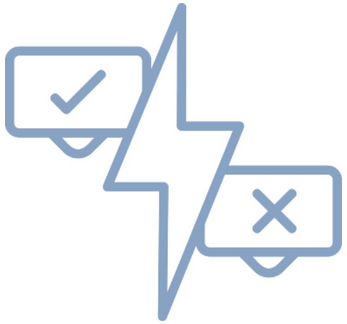
When a new VA business line wanted to integrate with VA Profile, the process was:

1. Partner contacts VA Profile team in one of several ways
2. VA Profile staff manually assesses technical requirements
3. Custom development work begins (6-12 month timeline)
4. Testing happens in production, creates data quality issues
5. No formal handoff, partner goes live without clear protocols

With a playbook, each integration was a new process.

Result: Integration timelines stretched 12-18 months while partners were not told. VA Profile staff got used to creating custom solutions.

2) Data Conflicts with No Resolution Process



Scenario: Veteran updates their phone number in the VA.gov mobile app. But the VHA (healthcare) system has a different number on file. Which is correct?

VA Profile had no standardized data validation rules or conflict resolution protocols. When systems disagreed:

- Some overwrote veteran-entered data
- Some created duplicate records, a data integrity issue
- Veterans received conflicting information across channels

Result: 16 million veteran profiles, but inconsistent data reliability.

3) Backlog of Integration

A small team handled integration requests, meaning prioritizing onboarding for a partner was arbitrary, or even got lost among other Jira tickets.

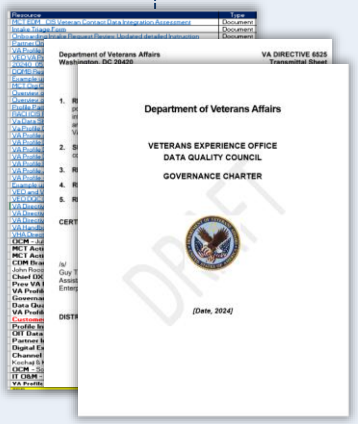
"Sometimes you open a ticket and realize it's from several months ago, and the partner is missing key information to move forward with their request. It can be hard tracking down the missing information, and the partner may even forget they did this at this point."

Result: A never-ending backlog of requests and a constant goose-chase for information out-of-sync with partner timelines.

Veterans Affairs: VA Profile

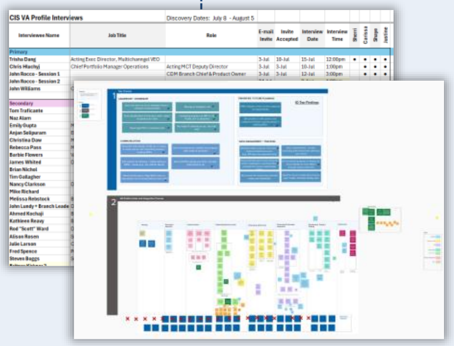
2. Define

Background Research
Research + review existing artifacts and materials, industry best practices



Best Practices:
- Business / IT Process Integration
- Data Management
- Partner Experience
- Knowledge Management
- Change Management

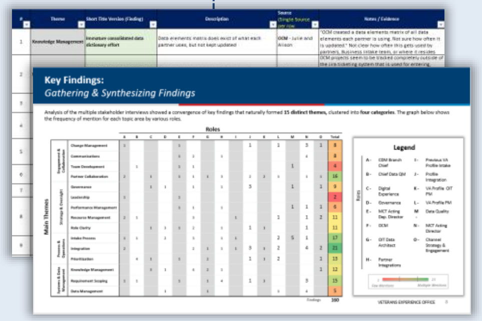
Data Collection
Functional interviews + whiteboarding sessions



18 Interviews
27 Key Stakeholders
31 Artifacts Reviewed

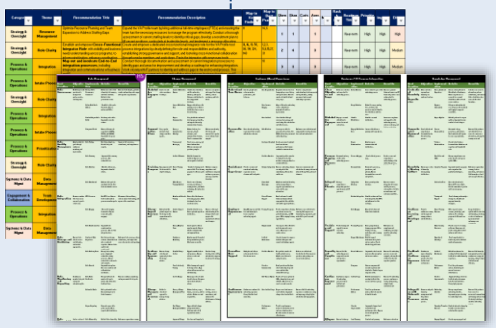
3a. Develop

Analyze + Synthesize Key Findings
Qualitative methods to identify valuable insights + common themes



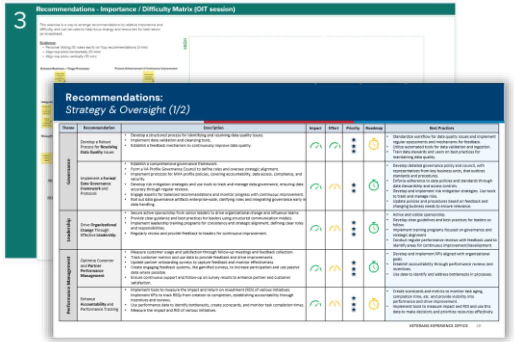
Validity + Priority Whiteboarding Sessions with key stakeholders

Develop Actionable Recommendations
Summarize key findings, align themes with objectives, prioritize themes and create recommendations



4. Deliver

Communicate Top Recommendations
Apply prioritization framework to rec's, map to best practices, create implementation timeline



150+ insights clustered into **15 common themes,** with top **36 recommendations**, informed by best practice research across 5 main areas.

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3b. Develop – Summary of Current VA Profile Intake and Integration Process

Current State Process Map (high level)

6 phases work in conjunction to continuously progress and develop the VA Profile for implementation while meeting regulatory requirements.

Identify VA Profile requirement

Awareness

Gather and document business requirements and needs related to the VA Profile Intake and Integration.

Objectives:
- Facilitate stakeholder engagement in defining business needs
- Verify use cases align with business requirements

Business Intake

Stakeholders are informed about the VA Profile Intake and Integration process, its importance, and their roles within it.

Objectives:
- Increase understanding + visibility of process
- Foster culture of proactive engagement + participation
- Comm. to all stakeholders process benefits+ impacts

Data Intake

Identify, assess, and integrate technological solutions that support VA Profile function and implementation.

Objectives:
- Validate data and user stories for security implications
- Implement and configure technology solutions effectively

Technology Intake

Collect, validate, and integrate data necessary for the VA Profile Intake and Integration process.

Objectives:
- Verify data accuracy, completeness, and consistency
- Enact data governance + QC measures

Development, Testing, Fielding

Maintain and continuously improve the VA Profile Intake and Integration process over time

Objectives:
- Monitor + evaluate process performance
- Request Partner feedback for continuous improvement

Continuous Sustainment

Develop, test, and validate the user stories and processes required to integrate VA Profile.

Objectives:
- Address and resolve any issues or defects identified during testing
- Conduct rigorous testing to verify quality and performance

Continuous Strategy Oversight and Communications across processes

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Veterans Affairs: VA Profile

3. Synthesis – Making Patterns Visible

Qualitative analysis clustered insights into actionable themes:

Analysis Method:

- Coded interview transcripts and artifacts for recurring pain points
- Identified 150+ discrete insights
- Clustered into 15 common themes using affinity mapping
- Mapped themes to 5 strategic areas based on best practice research:
    1. Business/IT Process Integration
    2. Data Management
    3. Partner Experience
    4. Knowledge Management
    5. Change Management

Best Practice Research:

Research on high-performing platform organizations (AWS, Salesforce, Azure) handle partner onboarding found 4 common patterns:

1. Self-Service Playbooks: Partners can start integration without hand-holding.

2. Tiered Support Models: Clear escalation paths for technical vs. business issues

3. Automated Testing Environments: Sandbox for partners to test before production

4. Partner Success Metrics: Track adoption, satisfaction, time-to-value

These became our North Star for recommendations.

4. Recommendations – Prioritized for Action

We developed 36 specific recommendations, then designed a prioritization framework to help VEO decide what to implement first.

Prioritization Framework (Impact vs. Effort Matrix):

For each recommendation, we assessed:

- **Potential Impact:** High/Med/Low Effect on trust, adoption, efficiency
- **Level of Effort:** Resources, time, organizational change required
- **Timeframe:** Quick Win (0-3 months), Medium (3-6 months), Long-term (6-12 months)

Validation Method:
To reveal the top rec's, I facilitated an Importance-Difficulty exercise with key stakeholders:

Workshop Design:

- Presented all 36 rec's without prioritization
- Asked stakeholders to rate:
    - "How important is this to achieving 90% trust?" (1-5 scale)
    - "How difficult is this to implement?" (1-5 scale)
- Discussed outliers
- Built consensus on top 10 priorities

This collaborative approach ensured buy-in. Stakeholders owned the priorities because they helped set them.

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Veterans Affairs: VA Profile

Deliverables

35-Page Strategic Assessment

Comprehensive analysis including:
- Current State process map + pain point analysis
- 150+ synthesized insights in 15 themes
- 36 prioritized recommendations with roadmaps
- Best practice research across 5 areas
- ROI projections for top recommendations

Integration Playbook

Practical guide for partner onboarding:
- Step-by-step integration process (what, when, who's responsible)
- Requirements templates + checklists
- Communication protocols + escalation paths
- Testing protocols for data validation
- Success metrics + feedback mechanisms

Projected Impact on VA Profile Operations:

IMMEDIATE (Within 3 months):
- RACI Matrix Implementation - reduced stakeholder confusion, clear escalation paths
- Standardized Testing Protocols - all new integrations now use sandbox before production

LONG-TERM (Strategic positioning):
- VA Profile positioned for expansion from 40 to 100+ partners
- Scalable operations model reducing dependency on individuals
- Foundation for achieving 90% veteran trust through reliable data

What I Learned:

Designing for Invisible Infrastructure
VA Profile is critical but invisible. I learned to articulate the human impact of backend systems: *when address data syncs correctly, a veteran gets their medication on time*. Making that connection helps explain to stakeholders why we should prioritize simple process improvements.

Prioritization is a Design Decision
Our team uncovered so many recommendations from our many interviews that it was hard to whittle them down. But the Importance-Difficulty exercise forced trade-offs and created stakeholder ownership. We knew that to give true direction, we had to triage the recommendations, since the top recommendations are the ones they'll actually implement.

Playbooks as Change Management Tools
VEO requested a playbook to have documentation for future training, but we also saw it as an opportunity to solidify our rec's. By classifying "how we do integrations," VEO culture could begin to shift from custom to standardized, from reactive to proactive.

Next Time:

Quantify Trust Metrics Earlier
The 'capturing 90% of veteran's trust' goal was motivating, albeit vague. If we'd defined measurable proxies for trust (such as data accuracy rate, partner satisfaction scores, or time-to-resolution) earlier, rec's could have been more explicitly tied to the goal.

Engage Veterans Directly
We interviewed internal staff and partners, but not veterans themselves. Including veteran voices on how data conflicts affected their experience would have added urgency and human stakes to backend process improvements.

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Boston Scientific Mobility Program

Design Problem:

While updating their design guidelines, Boston Scientific discovered their headquarters were heavily underutilized. Compounding on the popularity of their current work-from-home program, **a new program for "Agile Work"** - where employees will have unassigned seating when in the office - **was requested, as well as a $3 million renovation to support the program and refresh the space.**

Solution:

Based on research identifying potential for seat leveraging, employee badge data illustrating workplace utilization, and Boston Scientific's renovation constraints, an Agile Work program and accompanying redesign of the work floor were created for the IT team.

Overview:

User groups were identified:

- Workplace Functions
- Specialty Functions
- Workplace + Specialty

Data utilized:

- Previous Pilot Data
- Site Visits
- Interview + Surveys
- Best Practice Research
- Phone calls + e-mails
- Follow-up Visits

Primary issues identified:

- Low Utilization
- Outdated Tech
- Uninspiring Space
- Visual + Acoustical Privacy

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Boston Scientific Mobility Program

Strategy Process Steps: Define goals, data analyses, user analysis, space analysis, space redesign

Define Goals — **Data Analysis** — User Analysis — Space Analysis — Space Redesign

Define Goals

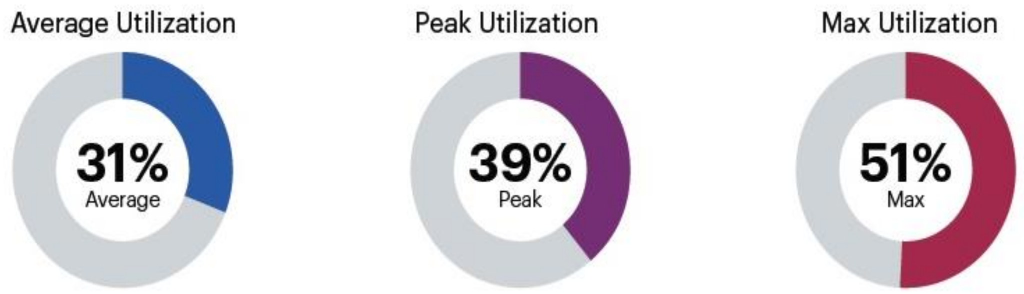
Workshop with leaders to define redesign goals and firm priorities

- ↑ collaboration + productivity
- Adapt to new paradigm of work
- Promote core values
- Attract + retain talent
- Increase cost-efficiency

Data Analysis

Data collected from real estate, peers, a pilot study, and leaders

Space Utilization Data
↓
Benchmarking Pharma Peers
↓
Data Synthesis of Previous Pilot
↓
Business Unit Interviews



On average:

50% **of employees are in office less than 3.5 days per week**

456 **workstations are empty every day** of the business week at Marlborough

- Passive quantitative metrics, like badge data, allowed us to track office on-site utilization only by day
- Qualitative research was based on surveys from previous pilot, business unit interviews, and feedback from employees in townhalls

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Boston Scientific Mobility Program

Define Goals — Data Analysis — **User Analysis** — Space Analysis — Space Redesign

User groups were evaluated for Agile Work by anticipating spatial needs on- and off-site

A. User Profiles:

Workplace Center Functions | **Specialty Center Functions** | **Both Workplace + Specialty Functions**

B. Agile Work candidacy considerations

User profiles evaluated based on:

- ❑ Paperwork, tech, phone usage
- ❑ Need for on-site resources
- ❑ Accessibility / collaboration
- ❑ Level of confidentiality

➢ **Low Agile Work Potential:**

- Need to be tethered to specific desk or bench
- Frequent need for specific equipment or resources that are only available in-office
- Desire for physical adjacency to team(s) or accessibility

➢ **High Agile Work Potential:**

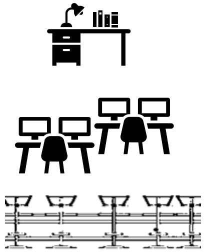
- High locational flexibility
- Low need for workplace spaces or team adjacencies
- Low need for heavy technology, storage, or physical documentation

The 'Agile Work' Program

Classifications

Boston Scientific wanted to see if IT would be good candidates for an 'agile work' program that redistributed office real estate based on needs of the job description, allowing employees to have seating in:

1. **Permanent desk / office** – belongs to one user
2. **Group addresses** – unassigned seating with similar user types
3. **Touchdown benching** – unassigned benching for anyone: contractors, visitors, etc.



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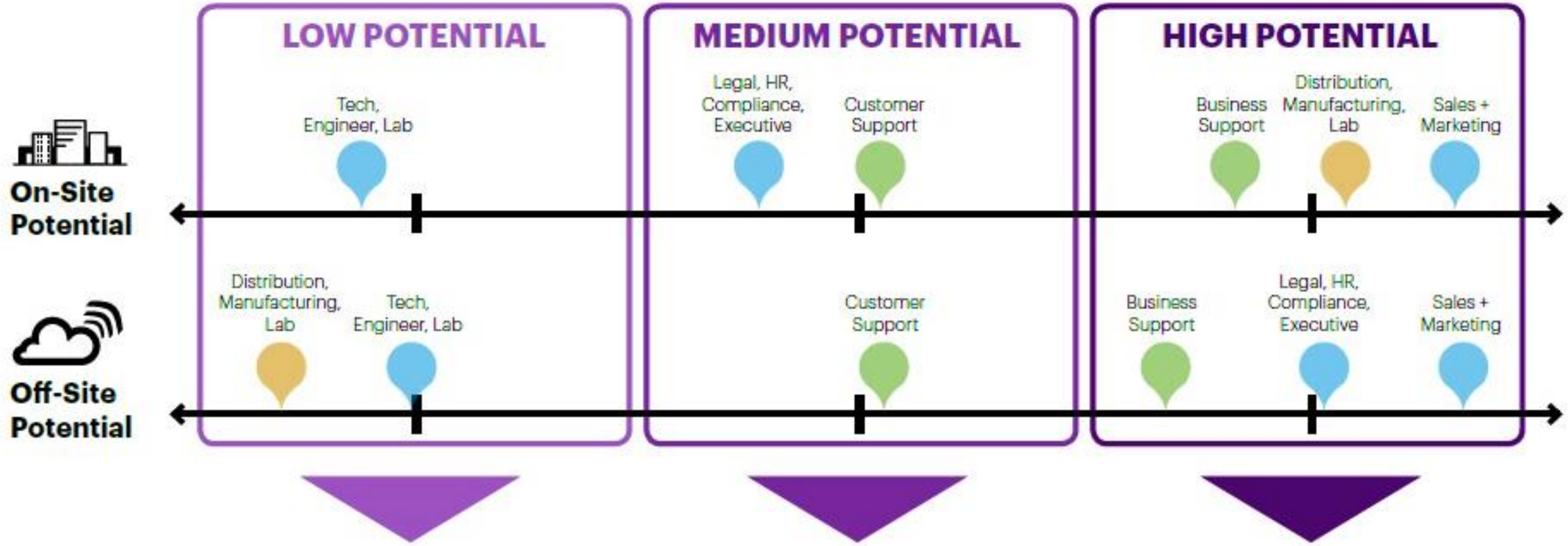
Boston Scientific Mobility Program

Define Goals — Data Analysis — **User Analysis** — Space Analysis — Space Redesign

User groups were evaluated for Agile Work by anticipating spatial needs on- and off-site

Agile Work Candidacy Spectrum



Spatial Implication:

CASE-BY-CASE LIFESTYLE CHOICE

- Dedicated desk or touchdown
- Adjacent to similar User Profiles

GROUP ADDRESS (NEIGHBORHOOD)

- Unassigned seat with touchdown
- Adjacent to similar User Profiles

CENTRALLY-LOCATED TOUCHDOWN

- Unassigned seat anywhere

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Boston Scientific Mobility Program

Define Goals — Data Analysis — User Analysis — **Space Analysis** — Space Redesign

Based on 'Business Support' workers' high potential to adopt Agile Work and 31% office utilization, an agility program was created for the IT team

IT population: breakdown + anticipated growth

2016

Existing Distribution	Target Ratio / Quantity
Employees	125 people
Contractors	69 people
Existing Population	194 people

~2020

Projected Distribution	Target Ratio / Quantity
Growth Rate	5%
Time Period	3-5 years
Employees	131 people
Contractors	73 people
Estimated Population	204 people

New 'Agile Work' program: assumptions + seat leverage

Assumed Enrollment Distribution

Employees:

Non-Employees:

Assume all Workflex participants plus an additional 10% will enroll

30% employees in Workflex
80% non-employees

Population Distribution	Estimated Enrollment
Employees	39 participants
Contractors	58 participants
AgileWork Population	97 participants

1 flex station: 1.80 AgileWork participants

*Seat leverage ratio built from an average of the Agility Assessment benchmarked range (1:1.1-1:3.0) and Spencer pilot goal (1:2.0)

Flex Seat Demand	54 flex seats

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Boston Scientific Mobility Program

Define Goals — Data Analysis — User Analysis — Space Analysis — **Space Redesign**

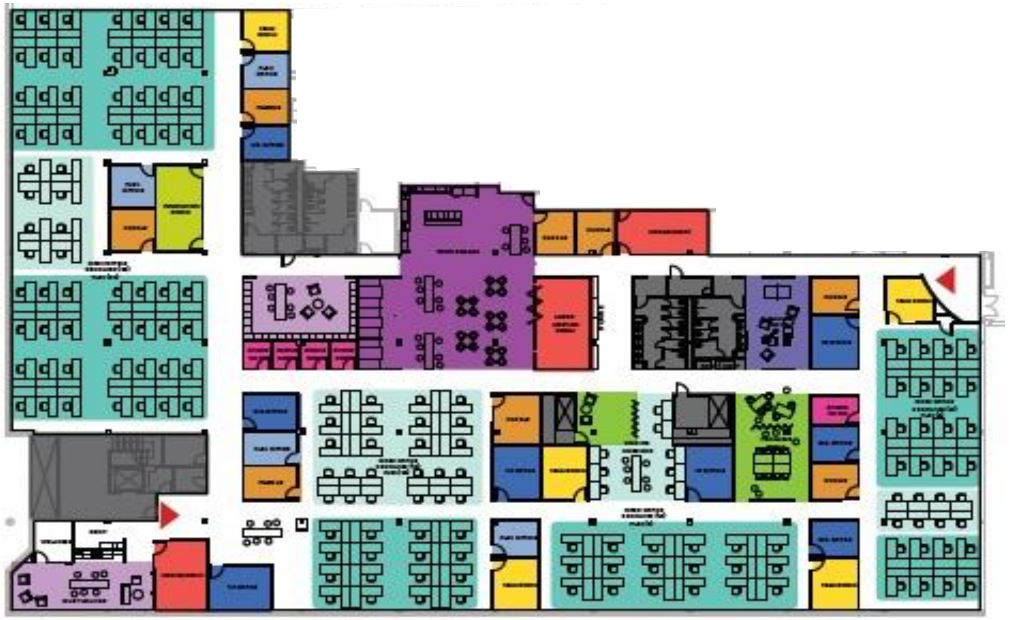
Work floor was remodeled, heeding Agile Work program requirements and budget

Design Problems identified in walkthroughs + interviews:

1. **Project Room:** Isolated, divides team + bars natural light
2. **Team Rooms** underutilized: poor layout + poor lighting
3. **Town Square + Huddles** underutilized: poor adjacencies + lack of technology / amenities

MY ROLE: As Design Strategist, I worked with my Manager, Architect, and fellow Strategist on a variety of projects. I conducted benchmarking, best practice research, user interviews, workshop facilitation and data synthesis / analysis for our clients, as well as designing deliverables and leading interns in tool kit developments.

Design Solution, new collab spaces + seat sharing:



Space Type

Point of Entry	Huddle Room
VP Office	Team Room
Director Office	Medium Conf. Room
Flex Office	Large Conf. Room
Standard Workstation	Innovation Room
Flex Desk	Hackable Zone
Flex Bench	Town Square
	Community Hub
	Game Space
	Phone Room

New collab spaces:
- More Team rooms
- 2 Innovation rooms
- Community Hub
- Game space
- Expanded Town Square

Seat sharing configuration:
- flexible workstations for 'agile' employees

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To & From App

IDEO Make-a-thon – Placed in Top 3

Design Problem:

Bitcoin and Fidelity are looking to discover the next generation's solutions to transactions, participation, and trust among people by leveraging emerging technologies. **Explore the intersection of technology and human needs to identify new market spaces and prototype venture solutions. Plan and prototype a starter kit for transactions among peers.**

Solution:

Digital prototype of a new app for students was created. App addresses imbalances in social interactions by logging events as memories and gratitude as the method of reimbursement. App includes virtual gestures as well as offerings from local businesses in array of appreciative deeds available to users.

Overview:

User profiles were created:

- Takers
- Helpers
- Forgetfuls
- Tutors

Data collected over 3-day period:

- Journey Mapping
- Traces + Observations
- Interviews
- Text & e-mail surveys

Primary issues first identified:

- Reciprocity
- Accurate Records
- Accountability

Primary issues identified after **data:**

- Gratitude
- Connection
- Partnerships

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To & From App

IDEO Make-a-thon – Placed in Top 3

1. User Profiles were created for transactions and borrowing among peer groups:

Taker Ted

borrows but never returns favors

FEEL: Expectant of others to help him but doesn't feel obligated to reciprocate

DO: Never has any money on him, always has others pay

SAY: "I forgot my wallet… thanks for covering again!"

Helper Hannah

will do any favor asked

FEEL: Likes /doesn't mind helping others, not comfortable asking for return favors

DO: Routinely doing favors for others without protest

SAY: "Oh, sure, whatever you need."

Forgetful Fay

wants to reciprocate but forgets deeds others do

FEEL: Guilty when doesn't remember if she owes friend

DO: Accidentally has others pitch in more than she intended

SAY: "Did you get the tickets last time or did I?"

Tutor Tina

likes to help others succeed

FEEL: Confident in her own abilities and wants to help others reach goals

DO: Offers time for homework help, leads study groups

SAY: "Tomorrow we can go over the problem set after class."

What would incentive each one?

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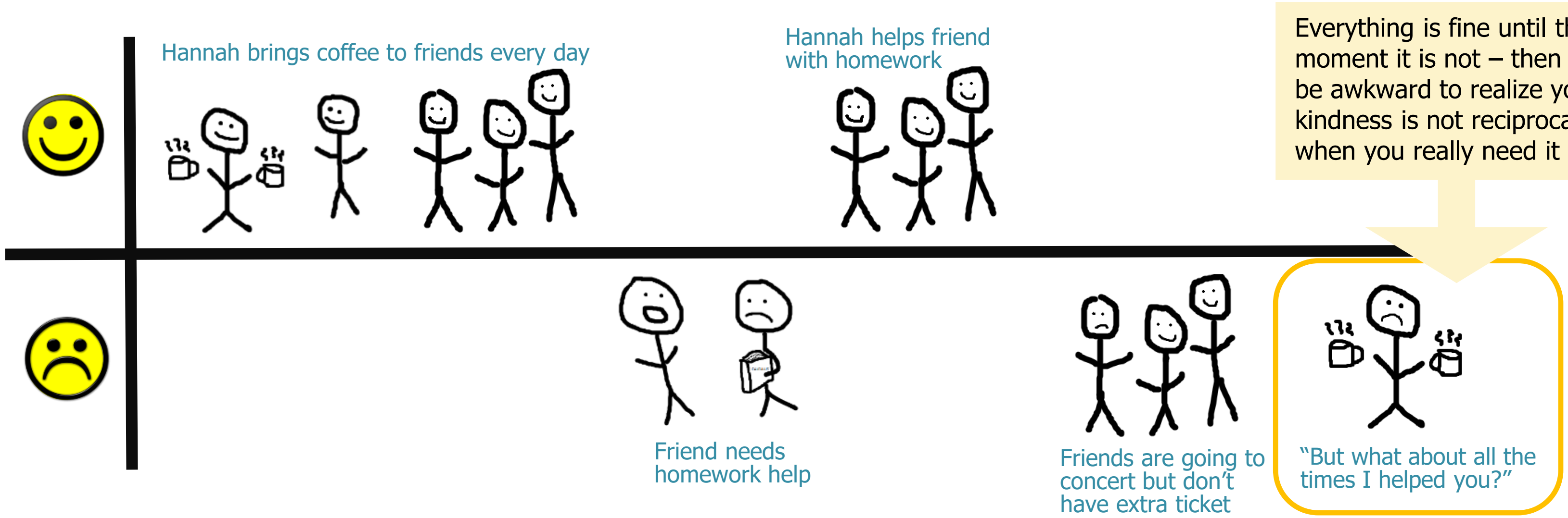
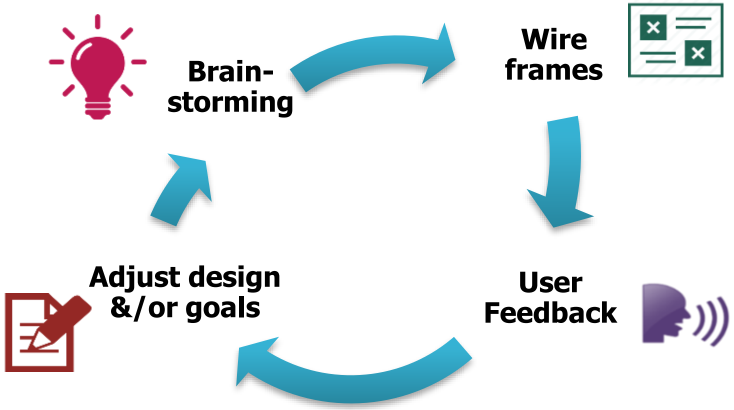
2. User Journeys were explored to see where the pain points could be:



After examining user profiles and user journeys, we decided we wanted an app to reduce the awkwardness of giving / returning favors between friends (see *Fig 1*). **We wanted to create a way to hold friends accountable, with a reward system for reciprocating favors.**

3. Design Process:



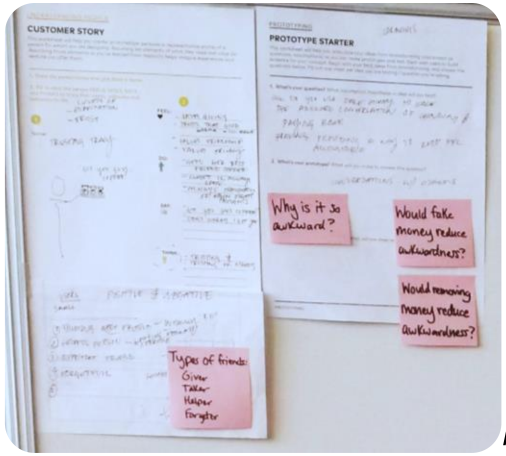


Fig 1

Creation: paper mock-ups of an app that keeps count of the favors a network of people perform (see *Fig 2* & 3)

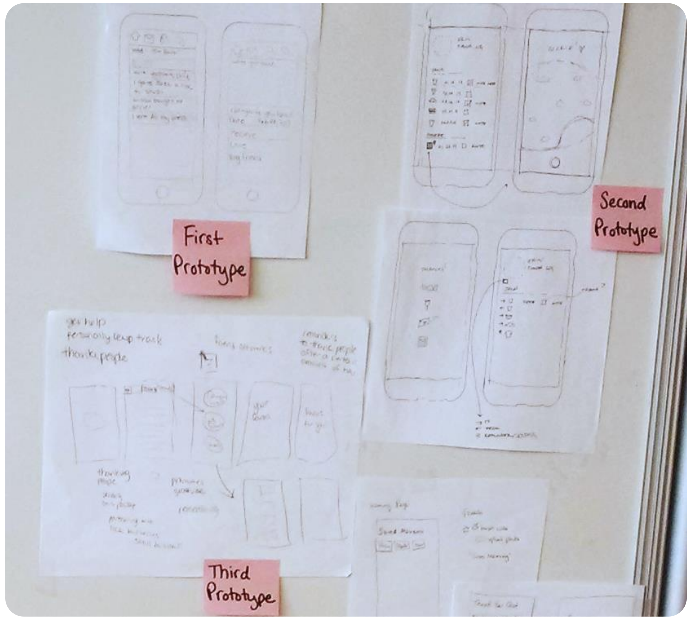


Fig 3

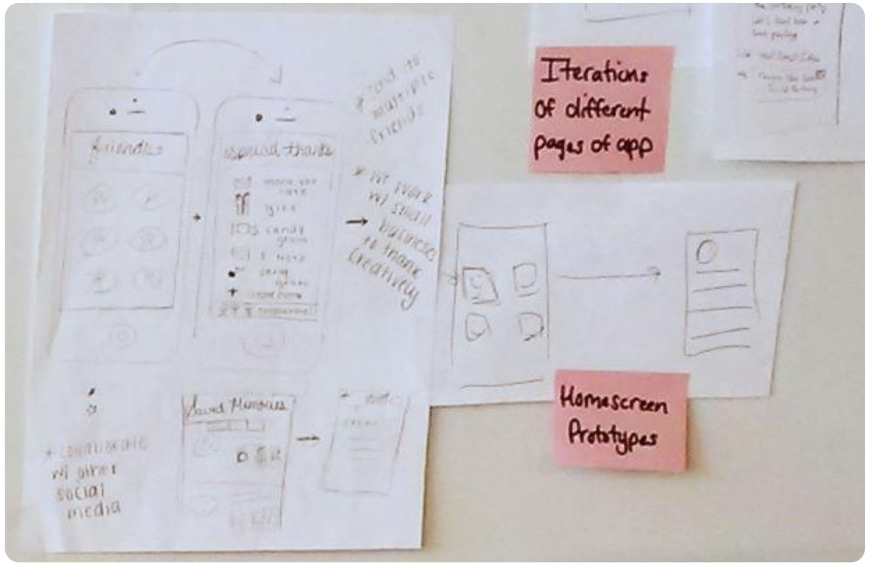


Fig 4

User Feedback: research via interviews and surveys

1) **Interviewed peers** on what was not working, what was missing, and what parts of the app would they not use:

- App was easy but needed to be more fun
- Liked how app helps them remember who did favors
- Don't want app to make them feel like a bad friend
- Unsure if favor records are accurate or socially acceptable

2) **Surveyed peers** via e-mail and text, asking what kind of reciprocity they expected after doing a favor:

- Most people told us a "simple thank you is enough"
- "Few people bother to say 'thank you' these days"

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To & From App

After creating wireframes and interviewing students on their feedback, we found that most people were not concerned with a tit-for-tat friendship. They did not have a problem with uneven lending among their friends; they only cared about feeling appreciated with a simple "thank you." We realized we had assumed the negativity of a pain point came from the lack of reciprocity, when in fact, the overwhelming feedback was that a 'thank you' was all a person wanted from their friend. We decided to shift the focus of our app and design around the concept of gratitude.

Goal Adjustment based on feedback – our app would handle reciprocity by focusing on gratitude

Gratitude reduces stress and improves:

- grades
- immunity
- sleep patterns
- teamwork
- feelings of belonging
- social networks
- relationships

Chose two benefits of gratitude to focus and reclarify goals of product:

WHO we design for	Students age 18-30's, tech-savvy students / young adults
What we want to LEARN	How can we reduce awkwardness of reciprocity in friendships through showing gratitude?
What we're MAKING	An app that makes gratitude social and incentivizes people to track favors - monetary, emotional, educational, practical, etc. - through thanking friends

Brainstorm: what gestures could thank friends and what student groups / businesses could assist in these gestures?

- How might we utilize local resources?
- How might we weave it into the design of the app?

Prototyped physical thank you cards and candy boxes for 'candy grams' (see *Fig 4* and *5*)

Fig 5

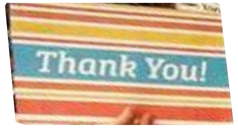

Fig 6

MY ROLE: Our team of 4 consisted of a RISD design student, an HBS student, and a psychology student. I was the human factors grad student, servicing the team in user personas, journey mapping, user interviews, prototype design and testing, and presentation.

To & From App

Outcome: Digital prototype was constructed in Illustrator.

To&From App allows people to customize their own list of favors (given and received) among their friend group(s), choose both physical and virtual ways to thank others, and log memories in a quick, easy, and fun way. While initially virtual, the app can partner with local businesses to offer creative ways to show appreciation for others, from hand-written notes to cards, gifts, and local collaborative events.

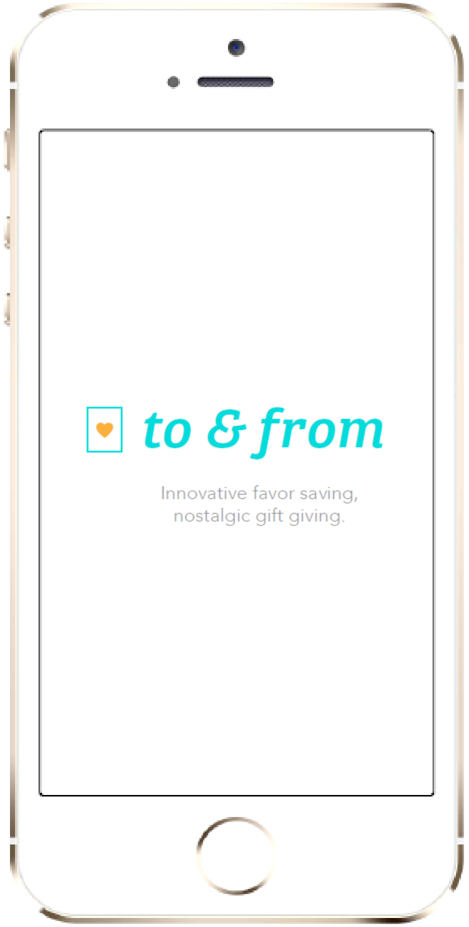

Fig 6:
Start Screen

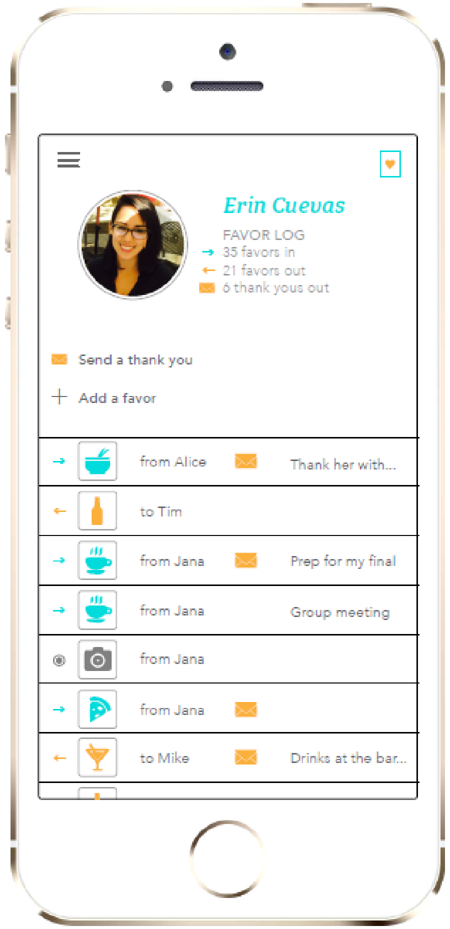

Fig 7:
Favor log screen

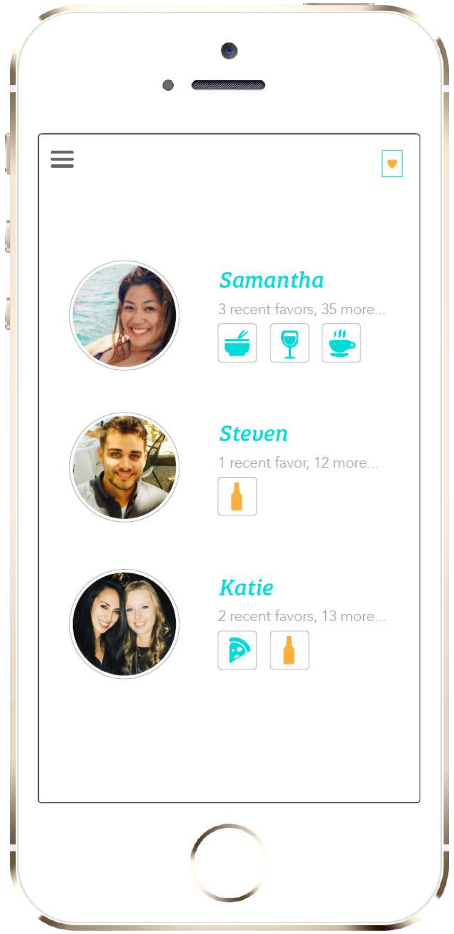

Fig 8:
List of friends screen

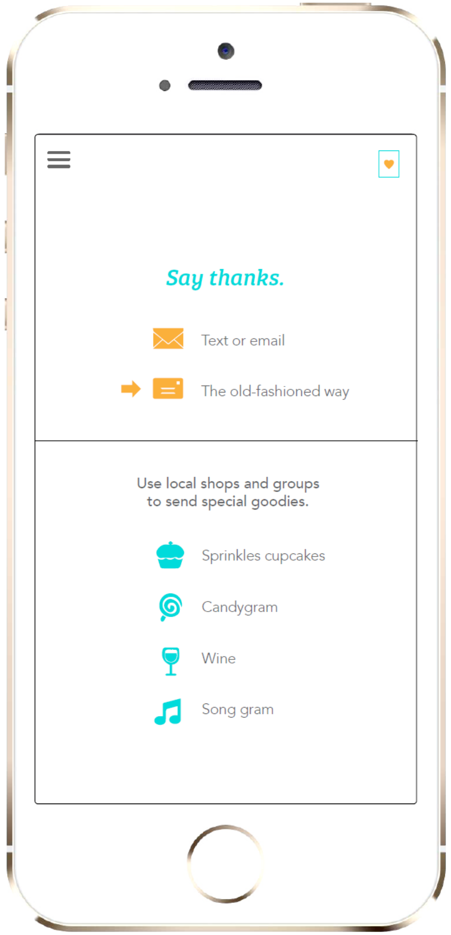

Fig 9:
'Thank You' options screen